MIL-C-24105C(SH)
28 September 1984
SUPERSEDING
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8 June 1979
(See 6.5)

MILITARY SPECIFICATION

CONVERTERS, SYNCHRO SIGNAL (PANEL MOUNTED)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for synchro signal converters (hereinafter referred to as converter) used for increasing the capacity of a synchro signal transmission system, converting from 60 to 400 hertz (Hz) synchro transmission or from 400 to 60 Hz synchro signal transmission, converting relative bearing signals to true bearing signals, to change synchro speeds, and to isolate reflected oscillations from the input synchro signal bus.

1.2 Classification. Converters shall be of the following type designation, as specified (see 6.2.1):

A	1	B	C
Number and gear ratio of input and output channels (see 1.2.1)	Accuracy of 1/18X output (see 1.2.1.1)	Input and output voltages and frequencies (see 1.2.2)	Size and type of output synchros (see 1.2.3)

1.2.1 Number and gear ratios of input and output channels. The number and gear ratios of the input and output channels shall be identified as follows:

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6125

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A Single input - single output - one channel - direct coupled - (see figure 1).
B Single input - single output - two independent channels - direct coupled - (see figure 1).
C Single input - single output - three independent channels - direct coupled - (see figure 1).
D Single input - single output - one channel - 2 to 1 gear ratio - (see figure 2).
E Single input - single output - two independent channels - 2 to 1 gear ratio - (see figure 2).
F Single input - single output - one channel - 1 to 2 gear ratio (see figure 2).
G Single input - single output - two independent channels - 1 to 2 gear ratio (see figure 2).
H Single input - triple output - 1 to 1/18, 1 and 36 gear ratio - (see figure 3).
I Single input - triple output - 36 to 1/18, 1 and 36 gear ratio - (see figure 4).
J Double input - triple output - 1 and 36 to 1/18, 1 and 36 gear ratio - (see figure 5).
K Double input - double output - 1 and 36 to 1 and 36 gear ratio - (see figure 6).
L Double input - triple output - 1 and 36 to 1/18, 1 and 36 gear ratio - (see figure 7).

The number of input and output channels also determine the number of input and output devices.

1.2.1.1 1/18 speed output accuracy. The accuracy of the 1/18 speed output of triple output converters shall be identified as follows:

1 0.2 degree accuracy
2 0.5 degree accuracy
(see figures 3, 4, 5 and 7)

1.2.2 Input and output voltages and frequencies. The input and output voltages and frequencies shall be identified as follows:

A 115 volts (V) - 60 Hz input - 115 V - 60 Hz output.
B 115 V - 60 Hz input - 115 V - 400 Hz output.
C 26 V - 60 Hz input - 26 V - 60 Hz output.
D 26 V - 60 Hz input - 115 V - 400 Hz output.
E 115 V - 400 Hz input - 115 V - 60 Hz output.
F 26 V - 400 Hz input - 115 V - 400 Hz output.
G 115 V - 400 Hz input - 26 V - 60 Hz output.
H 26 V - 400 Hz input - 115 V - 60 Hz output.
I 26 V - 60 Hz input - 115 V - 60 Hz output.
J 115 V - 400 Hz input - 115 V - 400 Hz output.
K 115 V - 60 Hz input - 26 V - 60 Hz output.
L 26 V - 400 Hz input - 26 V - 60 Hz output.
M 115 V - 60 Hz input - 26 V - 400 Hz output.
N 115 V - 400 Hz input - 26 V - 400 Hz output.
O 26 V - 60 Hz input - 26 V - 400 Hz output.
P 26 V - 400 Hz input - 26 V - 400 Hz output.

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1.2.3 Size and type of output synchros. The size and type of output synchros shall be identified as follows:

A	23CX	- Synchro control transmitter.
B	31CX	- Synchro control transmitter.
C	37CX	- Synchro control transmitter.
D	23CDX	- Synchro control differential transmitter.
E	37CDX	- Synchro control differential transmitter.
F	23TRX	- Synchro torque receiver-transmitter.
G	31TRX	- Synchro torque receiver-transmitter.
H	37TRX	- Synchro torque receiver-transmitter.
I	23TDX	- Synchro torque differential transmitter.
J	31TDX	- Synchro torque differential transmitter.
K	37TDX	- Synchro torque differential transmitter.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

FF-B-171	- Bearings, Ball, Annular (General Purpose).
FF-B-185	- Bearings, Roller, Cylindrical; and Bearings, Roller, Self-Aligning.
FF-B-187	- Bearings, Roller, Tapered.
QQ-A-200	- Aluminum Alloy, Bar, Rod, Shapes, Structural Shapes, Tube, and Wire, Extruded; General Specification for.
QQ-A-225	- Aluminum and Aluminum Alloy Bar, Rod, Wire or Special Shapes; Rolled, Drawn, or Cold Finished; General Specification for.
QQ-A-250	- Aluminum and Aluminum Alloy Plate and Sheet; General Specification for.
QQ-A-591	- Aluminum Alloy Die Castings.
QQ-A-596	- Aluminum Alloy Permanent and Semipermanent Mold Castings.
QQ-A-601	- Aluminum Alloy Sand Castings.
QQ-C-320	- Chromium Plating (Electrodeposited).

MILITARY

MIL-I-10	- Insulating Materials, Electrical, Ceramic, Class L.
MIL-I-631	- Insulation, Electrical, Synthetic-Resin Composition, Nonrigid.
MIL-S-901	- Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-C-5541	- Chemical Conversion Coatings on Aluminum and Aluminum Alloys.

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MILITARY (Continued)

MIL-R-6130 - Rubber, Cellular, Chemically Blown.
MIL-T-7928 - Terminals, Lug: Splices, Conductor: Crimp Style, Copper, General Specification for.
MIL-Q-9858 - Quality Program Requirements.
MIL-P-10971 - Pin, Spring, Tubular (Coiled and Slotted).
MIL-F-15160 - Fuses: Instrument, Power, and Telephone.
MIL-F-15160/77 - Fuses (Indicating), Style F77A.
MIL-T-16366 - Terminals, Electrical, Lug and Conductor Splices, Crimp-Style.
MIL-W-16878 - Wire, Electrical, Insulated, General Specification for.
MIL-W-16878/4 - Wire, Electrical Polytetrafluoroethylene (PTFE) Insulated, 200°C, 600 Volts, Extruded Insulation.
MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Repair Parts: Packaging and Packing of.
MIL-B-17931 - Bearings, Ball, Annular, For Quiet Operation.
MIL-F-19207 - Fuseholders, Extractor Post Type, Blown Fuse Indicating and Nonindicating, General Specification for.
MIL-F-19207/38 - Fuseholders, Extractor Post Type, Blown Fuse Indicating, Type FHL57G.
MIL-S-19500 - Semiconductor Devices, General Specification for.
MIL-R-20092 - Rubber Sheets and Assembled and Molded Shapes, Cellular, Synthetic, Open Cell (Foamed Latex).
MIL-S-20708 - Synchros, General Specification for.
MIL-A-21180 - Aluminum Alloy Castings, High Strength.
MIL-S-22432 - Servo Motors, General Specification for.
MIL-S-23190 - Straps, Clamps, and Mounting Hardware, Plastic for Cable Harness Tying and Support.
MIL-C-28748 - Connectors, Electrical, Rectangular, Rack and Panel, Solder Type and Crimp Type Contracts, General Specification for.
MIL-C-28748/9 - Connector, Electrical, Rectangular, Rack and Panel, Polarized Center Jackscrew or Guidepin Style, Crimp Type Removable Pin Contracts Size 16.
MIL-C-39022 - Capacitors, Fixed, Metallized, Paper-Plastic Film, or Plastic Film Dielectric, Direct and Alternating Current (Hermetically Sealed in Metal Cases), Established Reliability, General Specification for.
MIL-T-55164 - Terminal Boards, Molded, Barrier, Screw and Stud Types, and Associated Accessories, General Specification for.
MIL-S-81746 - Servtorq, General Specification for.

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STANDARDS

MILITARY

MIL-STD-167-1	- Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-242	- Electronic Equipment Parts Selected Standards.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-461	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-701	- Lists of Standard Semiconductor Devices.
MIL-STD-710	- Synchros, 60 and 400 Hertz.
MIL-STD-740	- Airborne and Structureborne Noise Measurements and Acceptance Criteria of Shipboard Equipment.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-965	- Parts Control Program.
MIL-STD-1303	- Painting of Naval Ordinance Equipment.
DOD-STD-1399, Section 300	- Interface Standard for Shipboard Systems, Electric Power, Alternating Current (Metric).

HANDBOOKS

MILITARY

MIL-HDBK-214	- Synchros.
MIL-HDBK-217	- Reliability Prediction of Electronic Equipment.
MIL-HDBK-225	- Synchros, Description and Operation.

2.1.2 Publication. The following publication forms a part of this specification to the extent specified herein.

PUBLICATION

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

DDS 430-2 - Procedure for Determining Synchro System Loading Capacity.

(Copies of specifications, standards, handbooks and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

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ASTM

A 153 - Zinc Coating (Hot-Dip) on Iron and Steel Hardware.
B 633 - Electrodeposited Coatings of Zinc on Iron and Steel. (DoD adopted)

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Qualification. The converters furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.3 and 6.3).

3.2 Description and operation.

3.2.1 Description. Converters shall consist of one or more input devices, an amplifier, a servo unit, and one or more output synchro devices as specified (see 1.2 and 6.2.1). Input devices shall be synchro control transformers (with capacitors and an amplifier unit if required) or SERVTORQ's (see 3.2.1.2), as specified (see 1.2.2 and 6.2.1). Servo unit shall be coupled between the input and output devices directly or by gearing (see figures 1 through 7). The above parts shall be assembled in a panel mounted enclosure as specified on figure 8.

3.2.1.1 Signal characteristics. The various signals or signal combinations that may be transmitted through a converter shall be identified as specified in 3.2.1.1.1 through 3.2.1.1.2.3.

3.2.1.1.1 Single channel converters.

3.2.1.1.1.1 Single channel converters that are direct coupled (see figure 1) shall receive and transmit one of the following signals:

(a) 360 degrees per revolution (Rev)
(b) 180 degrees per Rev
(c) 10 degrees per Rev
(d) 1 degree per Rev
(e) 2K yards per Rev
(f) 72K yards per Rev
(g) 1296K yards per Rev

3.2.1.1.1.2 Single channel converters with gearing (see figure 2) shall receive and transmit the following signals:

(a) 180 degrees per Rev input with 360 degrees per Rev output.
(b) 360 degrees per Rev input with 180 degrees per Rev output.

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3.2.1.1.2 Multiple channel converters.

3.2.1.1.2.1 Single input converters (see figure 3 or 4) shall receive and transmit one of the following combinations of signals:

(a) 360 degrees per Rev input with 10 degrees per Rev, 360 degrees per Rev and 6480 degrees per Rev outputs.
(b) 72K yards per Rev input with 2K yards per Rev, 72K yards per Rev, and 1296K yards per Rev output.

3.2.1.1.2.2 Double input - triple output converters (see figure 5 or 7) shall receive and transmit one of the following combinations of signals:

(a) 360 degrees per Rev and 10 degrees per Rev inputs with 6,480 degrees per Rev, 360 degrees per Rev and 10 degrees per Rev outputs.
(b) 72K yards per Rev and 2K yards per Rev inputs with 1296K yards per Rev, 72K yards per Rev and 2K yards per Rev outputs.

3.2.1.1.2.3 Double input - double output converters (see figure 6) shall receive and transmit one of the following combinations of signals:

(a) 360 degrees per Rev and 10 degrees per Rev.
(b) 180 degrees per Rev and 1 degree per Rev.
(c) 72K yards per Rev and 2K yards per Rev.

3.2.1.2 SERVTORQ. SERVTORQ is the Navy acronym for an integrated servo-mechanized torque receiver. It serves as a self-contained, remote angular positioning device for converting input synchro data into an accurate shaft position. The integrated assembly shall consist of a synchro control transformer (CT), a power supply, an amplifier, and a direct current (d.c.) torque or drive motor, all mounted in a size 23 synchro frame (see 3.2.1). The SERVTORQ shall have an output shaft, terminal connections, and operate in an ambient temperature of minus 67 degrees Fahrenheit (°F) to plus 185°F, and shall be in accordance with MIL-S-81746.

3.2.1.3 SERVTORQ nomenclature. The nomenclature 23ST64a shall designate a SERVTORQ (ST) with a maximum diameter of 2.3 inches (size 23) for nominal power supply frequencies of 60 or 400 Hz (64) at 115 V. The "a" following the frequency digits indicates the original or basic issue of the standard unit.

3.2.2 SERVTORQ functional details. Synchro control transmitter (CX) or torque transmitter (TX) (see figure 9) shall convert angular position data into a three-wire electrical signal, which shall excite the stator windings of the SERVTORQ CT via terminals S1, S2 and S3. CT rotor shall be mounted on the SERVTORQ output shaft. An error voltage, proportional to the difference between the transmitter (CX or TX) and SERVTORQ output shaft positions, respectively, shall be obtained from the rotor of the CT. The error voltage shall be injected into a phase sensitive demodulator (DEMOD), which shall convert the in phase alternating current (a.c.) error signal to a proportional d.c. level. The latter shall be amplified ("DC AMP" (see figure 9)) to drive the d.c. torque motor, which shall be mounted on the same shaft as the CT rotor. The phasing in the stabilization network shall be such that the error voltage shall be amplified to produce a torque on the rotor, which shall drive it to a null voltage position.

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At this point, the error shall be nulled, and rotation of the rotor and output shaft shall cease. A 1 to 1 correspondence between the transmitter input shaft position and the SERVTORQ output shaft position shall be achieved. Maximum permissible SERVTORQ loads for various converter configurations shall be as shown in table I. (See figures 1 through 7 for illustrations of SERVTORQ applications in synchro signal converters.)

TABLE I. Maximum permissible sizes of output synchros for various converter configurations.

Configuration	Reference figure number	Maximum load-output synchro(s)	
		Size	Types
Single channel SERVTORQ direct - coupled to output synchro	1	23, 31 or 37	TX, CX, CDX, TRX or TDX
Single channel 180 degrees per Rev input 360 degrees per Rev output	2	23	TX, CX, CDX, TRX or TDX
Two speed input (2 SERVTORQ and 2 CTs) - three speed output	7	23	TX, CX or TRX
Two speed input (1 SERVTORQ and 2 CTs) - three speed output	5	23	TX, CX or TRX
One speed input (1 SERVTORQ and 1 CT) - three speed output	3	23	TX, CX or TRX
One speed input - three speed output	4	23	TX, CX or TRX
Two speed input - two speed output	6	23	TX, CX or TRX

3.2.3 Operation. The typical cycle of operation of the converter shall be as follows: A synchro signal shall be transmitted to the synchro CT, or SERVTORQ, in the synchro signal converter. The signal received by CT shall be fed to the amplifier unit, where it shall be intensified and used to actuate the servomotor. The latter, through gearing or direct coupling, shall turn the CT rotor until the induced voltage is reduced to a null, thereby stopping the servo action. Simultaneously, the servomotor shall drive, directly or through gearing as required, the rotor(s) of the output synchro transmitter(s). The output synchros shall supply signals to remote synchro receivers, which shall correspond to the signal received by the CT(s).

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3.3 General features.

3.3.1 Characteristics not normally incorporated in the design of commercial equipment are required on Naval ships. The primary requirements which are the basis for the design of all Naval equipment shall be as follows:

(a) Maximum reliability for operation under shipboard conditions of heat, moisture, vibration, shock and ship motion as described herein.
(b) Material and design shall be lightweight and compact consistent with reliable shipboard operation.
(c) Maximum accessiblity for inspection and maintenance.
(d) Maximum simplicity of design consistent with other requirements described herein.
(e) Maximum resistance to corrosive action, in particular that of moist sea air and ultraviolet radiation.
(f) Maximum operating and maintenance economy.
(g) Maximum interchangeability of parts.
(h) Minimum number of necessary maintenance parts.
(i) Satisfactory identification of equipment and parts for ease in accurate requisitioning.
(j) Satisfactory drawings with complete and concise manufacturing and operating instructions.
(k) Maximum ease of repair by Naval personnel and with the facilities normally found on board ships.
(l) Maximum protection for operating and maintenance personnel.
(m) Maximum ease of installation.

3.3.2 Standard parts. Standard stock parts and hardware shall be used to as great an extent as practicable (see 3.6.2). For the purpose of this specification, standard stock is defined as material listed in the Federal Supply Catalog and includes such items as bearings, grease, oil, wire, bolts, screws, nuts and washers. Proprietary parts or parts available from one source shall not be used where equivalent or similar parts are available as standard parts or are available from two or more sources.

3.3.3 Interchangeability. Parts used in the equipment shall be interchangeable to the greatest extent practicable, in accordance with requirement 7 of MIL-STD-454.

3.4 Materials and processes. The following requirements in accordance with MIL-STD-454 shall be applicable to the materials and processes used in the manufacture of the converters described herein in addition to the requirements of this specification:

(a) Requirement 3 - Flammable materials.
(b) Requirement 4 - Fungus-inert materials.
(c) Requirement 11 - Insulating materials, electrical (see 3.5.5 and 3.6.11.2).
(d) Requirement 13 - Structural welding.
(e) Requirement 15 - Ferrous alloys, corrosion resistance (only austenitic stainless steel shall be used).

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(f) Requirement 17 - Printed wiring.
(g) Requirement 21 - Castings.
(h) Requirement 26 - Arc-resistant materials.
(i) Requirement 43 - Lubricants.

3.4.1 Toxic materials. Radioactive materials, mercury and heavy metal biocides shall not be used. Materials used in the construction of the converter shall not produce corrosive, deleterious or toxic fumes or vapors under all conditions herein.

3.4.2 Wood. Wood shall not be used.

3.4.3 Aluminum. Where the use of lightweight corrosion-resistant metal is desired, wrought aluminum alloys of the 5000 series shall be used insofar as practicable. These aluminum alloys shall be in accordance with QQ-A-200, QQ-A-225, or QQ-A-250. In general, aluminum alloy castings shall be in accordance with alloy 360 of QQ-A-591 or to alloy 356 of QQ-A-596 or QQ-A-601. High strength aluminum alloy castings shall be in accordance with alloy A 356 of MIL-A-21180.

3.4.4 Magnesium. Magnesium shall not be used.

3.4.5 Nonferrous materials except aluminum. Nonferrous materials, except aluminum, shall conform to commercial standards and shall be of a quality to meet the requirements of this specification.

3.4.6 Zinc. Zinc shall not be used where the increased electrical resistance of the surface, due to the protective treatment, has a deleterious effect on electrical performance.

3.4.7 Ceramics. Ceramics shall not be used for structural parts. Ceramics, where used for other applications, shall be in accordance with MIL-I-10.

3.4.8 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and shall be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.4.9 Painting. After machining, welding and brazing operations have been completed, the front panel of the converter enclosure shall have rust or other visible corrosion products removed and shall be thoroughly cleaned of grease, oil and dirt, and painted in accordance with the requirements of system 37 of MIL-STD-1303.

3.4.10 Protection against corrosion. In order to prevent corrosion, metal parts shall be of corrosion-resisting materials or other materials treated as specified in 3.4.10.1 to render them adequately resistant to corrosion. Internal parts fabricated of nonferrous materials contained in a watertight enclosure shall not be given a corrosion-resistant treatment.

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3.4.10.1 Corrosion-resisting treatments. The most applicable corrosion-resisting treatment of those listed below shall be applied after fabricating operations such as welding, machining, drilling and tapping have been completed. Corrosion-resisting treatments shall not be applied to surfaces which are in functional contact where gouging or binding may be a factor or where the treatment might interfere with normal functioning or where electrical grounding through the surface is required.

(a) Zinc coating, hot dip galvanizing in accordance with ASTM A 153, for parts other than threaded fasteners, unless part specifications contain other requirements.
(b) Electroplating of zinc in accordance with type II, Fe/Zn 13 of ASTM B 633, for surfaces not to be painted except that class 3 thickness may be used for externally threaded parts, bolts, studs and washers. Type II, class 1 plating may be used on parts which are continuously exposed to temperatures in excess of 150°F or are intermittently exposed for short periods to temperatures of approximately 300°F. Either type II, class 2, or type III, class 2 coating shall be used on surfaces that are to be painted.
(c) Electroplating of chromium in accordance with to QQ-C-320.
(d) Chemical film in accordance with class 1A of MIL-C-5541.

3.4.10.2 Selection of metals. In order to minimize corrosion attack due to electrolytic action between dissimilar metals when used in contact with each other, selection shall be in accordance with MIL-STD-889.

3.5 Parts and features - mechanical.

3.5.1 The following requirements in accordance with MIL-STD-454 shall be applicable to the mechanical parts and features of the converters:

(a) Requirement 12 - Fastener hardware (see 3.5.1.1).
(b) Requirement 21 - Castings.
(c) Requirement 63 - Special tools.

3.5.1.1 Fastener hardware. Thread cutting and self-tapping screws shall not be used. A lock washer shall be installed under the head of other than flat head machine screws. Flat head machine screws shall be secured with a staking compound.

3.5.2 Bearings. Bearings shall be in accordance with requirement 6 of MIL-STD-454 and as specified in 3.5.2.1 through 3.5.2.4.

3.5.2.1 Ball bearings. Ball bearings shall be in accordance with FF-B-171. They shall be applied and installed in accordance with the requirements of that specification. Grade 00 ball bearings shall be used wherever possible.

3.5.2.2 Roller bearings. Roller bearings shall be in accordance with FF-B-185 or FF-B-187 and shall be applied and installed in accordance with the requirements of these specifications.

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3.5.2.3 Sleeve-type bearings. Sleeve-type bearings shall not be used.

3.5.2.4 Noise tested bearings. Where required, noise tested bearings shall be in accordance with MIL-B-17931.

3.5.3 Rounded corners and edges. Corners and edges of external surfaces, normally exposed or to be painted, shall be rounded.

3.5.4 Protection of wiring. Edges of parts that will come into contact with wiring shall be rounded or provided with rounded insulating guards, grommets or pads to prevent damage to the internal wiring (see 3.6.11.4). Stand-offs or protective plates shall be provided where required to prevent wiring from being damaged by terminal nuts or screw threads. Insulating guards, grommets, pads, sheaths, stand-offs and protective plates shall be in accordance with type F, grade A of MIL-I-631.

3.5.5 Internal subassembly protection. Complete unit internal subassemblies shall be provided with means to prevent injury to pointers, dials and other parts when the subassembly is removed from its enclosure and rested on a work bench on either its top or bottom.

3.5.6 Drilled and tapped holes. Drilled and tapped holes shall be countersunk to remove burrs and sharp edges. Tapped holes in aluminum shall be fitted with helical inserts.

3.5.7 Size. Converter shall be as specified herein, and as shown on figure 8.

3.5.8 Mounting. Converter shall be mounted as shown on figure 8. Converter and front panel shall have the necessary strength and rigidity to prevent contact between the converter and the walls of the cubicle under the conditions of shock and vibration (see 3.7.12 and 3.7.14).

3.5.9 Handles. Two handles shall be provided. Handles shall be secured to, but detachable from, the front panel. Handles shall not extend more than 2-1/2 inches from the front panel (see figure 8). Figure 8 shall be used for guidance in the location of handles.

3.5.10 Accessibility. Parts and subassemblies of the converter shall be easily accessible for repair and adjustment. Adjustment of synchros shall not require any disassembly of the converter. Supporting structures shall be provided on the converter to allow repair work without having the unit resting on wiring, electrical connections, or parts that may be dislocated or damaged. Requirement 36 in accordance with MIL-STD-454 shall be used as a guidance with respect to accessibility.

3.5.11 Gearing. One gear train shall connect the servomotors, receivers, and transmitters. Gears shall provide smooth, continuous tooth contact without interferences, tight spots, loose spots or other irregularities. Every precaution shall be taken to reduce tooth pressure and backlash to a minimum. Gears fitted on shafts shall be secured thereto by means which will effectively prevent relative motion between the gear and the shaft (see 3.5.12). Gears shall be metallic.

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3.5.12 Shaft locking devices. Positive locking devices shall be used to secure collars, gears and similar parts to their associated shafts. Acceptable methods shall be in accordance with MIL-P-10971 or requirement 28 of MIL-STD-454. Taper pins shall not be used.

3.5.13 Weight. Weight of the converter shall be kept to a minimum.

3.5.14 Mechanical connections. Connections shall be supported in a manner which will preclude breakage and minimize changes in performance due to vibration, inclination, or shock encountered under the conditions of Naval service. Electrical (clamp contact) connections shall not depend upon wires, lugs, terminals, and so forth, clamped between a metallic member and an insulating pliable material. Connections shall be clamped between metal members (see 3.6.10).

3.6 Parts and features - electrical.

3.6.1 The following requirements in accordance with MIL-STD-454 shall be applicable to the electrical parts and features of the converters:

(a) Requirement 2 - Capacitors.
(b) Requirement 5 - Soldering (also covers mounting of electrical parts) (see 3.6.21.1.6).
(c) Requirement 14 - Transformers, inductors, and coils (see 3.6.10).
(d) Requirement 17 - Printed wiring.
(e) Requirement 18 - Derating of electronic parts and materials.
(f) Requirement 24 - Welds, resistance, electrical interconnections.
(g) Requirement 30 - Semiconductor devices.
(h) Requirement 33 - Resistors.
(i) Requirement 69 - Internal wiring practices (see 3.6.9).

3.6.2 Standard parts. Standard electronic and electrical parts, defined as those specified in MIL-STD-242, shall be used wherever practicable when the part is not specified herein. The procedure for the selection of electronic and electrical parts and materials not specified herein or in MIL-STD-242 shall be in accordance with MIL-STD-965.

3.6.3 Safety, ground potential and grounding. Safety and grounds shall be in accordance with requirement 1 of MIL-STD-454.

3.6.4 Electronic assemblies. Electronic assemblies shall be compact solid state units and shall not be potted. (Electronic assemblies internal to a device shall be potted, as required.) Plug-in type assembly devices are preferred and shall be used where practicable.

3.6.5 Fuses and fuseholders.

3.6.5.1 Fuses. Fuses shall be style F77A in accordance with MIL-F-15160 and MIL-F-15160/77 and shall be provided and installed as specified (see 6.2.1). Fuses under 1 ampere rating shall not be used. Fuse size shall be indicated by marking adjacent to each fuse.

3.6.5.2 Fuseholders. Fuseholders shall be type FHL57G in accordance with MIL-F-19207 and MIL-F-19207/38. The fuseholders shall be mounted on the front panel (see figure 8) or as specified (see 6.2.1).

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3.6.5.3 Blown fuse indicator circuit. Blown fuse indicator circuit used with type FHL57G fuseholders shall be energized as specified (see 6.2.1).

3.6.6 External electrical connections. External electrical connections to the converter shall be made by either a terminal board (type 10TB12 or 6TB24 in accordance with MIL-T-55164) or a connector (part number M28748/09F0N01A in accordance with MIL-C-28748 and MIL-C-28748/9) mounted on the rear of the converter (see 6.2.1).

3.6.7 Synchro signal selector. A synchro signal selector shall be provided in multispeed converters (for example, 1 and 36 speed; 1, 18 and 648 speed) in order to select the controlling signal from the input synchros. The synchro signal selector is called a speed switching network as shown on figures 5 and 6.

3.6.8 Power supply. Converters shall operate within the specified accuracy limits (see 3.7.1) from a type I shipboard 115 volts alternating current (Va.c.) power source, as specified in DOD-STD-1399, section 300. Power required within the amplifiers shall be obtained from internal sources energized from the synchro reference power supply. Rectification from a.c. to d.c. shall be accomplished by means of a full-wave bridge configuration. Transient protection shall be employed in the power supply design to prevent damage to the supply itself, as well as electronic assemblies powered by the supply (see 3.6.21.1.3 and 3.7.7).

3.6.9 Insulation resistance clearance. Clearances between any two electrical circuits or between any electrical circuit and ground (metal enclosure or chassis) shall be such as to meet the test conditions for insulation resistance specified in 4.6.9. Insulation resistance of the circuits test shall be not less than 10 megohms when tested as specified in 4.6.9. Creepage and clearance distances between conductors and other conductors, ground or frame in the equipment shall be in accordance with requirement 69 of MIL-STD-454 and in no case shall be less than necessary to prevent arcing or leakage during operation or service.

3.6.10 Transformers and inductors. Transformers and inductors shall be in accordance with requirement 14 of MIL-STD-454, grade 5, class S, life expectancy X, and shall be selected in accordance with MIL-STD-242. Power and audio input transformers shall have electrostatic shields, and the shields shall be grounded.

3.6.11 Wire. Internal hook-up wire shall be in accordance with MIL-W-16878 and MIL-W-16878/4.

3.6.11.1 The use of preformed cables and wiring harnesses is preferred to the point-to-point method of wiring.

3.6.11.2 Wiring shall be arranged to permit bundling by one or more of the following methods:

(a) Binding - Tape for binding shall be in accordance with requirement 11 of MIL-STD-454.
(b) Sleeving insulation - Sleeving insulation shall be in accordance with requirement 11 of MIL-STD-454.
(c) Tying - Plastic devices for tying shall be in accordance with MIL-S-23190. When the wire bundle is formed and secured, the device shall not loosen under vibration.

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3.6.11.3 Color coding and marking of wires shall be in accordance with requirement 20 of MIL-STD-454.

3.6.11.4 Wiring, both fixed and flexible, shall be protected to prevent chafing and abrasion due to vibration or flexing in accordance with 3.5.4. Plastic sheaths shall be in accordance with type C of MIL-I-631 and may be similar to a covering known as "zipper tubing". A sealing compound shall be used with this tubing. The use of clamps or wrapping and tying devices (see 3.6.11.2) with integral mounting facilities is preferred for securing fixed wiring. Metal clamps, if used, shall be insulated and shall not form a complete loop around the wiring.

3.6.12 Wire terminal lugs. Except for contacts used with connectors, pressure grip terminal lugs in accordance with figure titled "Type WTG terminals, round or rounded tongue, one bolt hole" of MIL-T-16366, or MIL-T-7928 shall be used. Insulated barrel terminal lugs shall be used. Terminal lugs with projections or "lips" shall be used in all cases where necessary to prevent turning of terminal lugs and shorting due to close clearances. Solder type terminal lugs may be used.

3.6.13 Printed wiring boards. Printed wiring or printed circuit boards shall be in accordance with requirement 17 of MIL-STD-454. Boards shall permit replacement of parts without special tools or techniques (see 3.6.21.1.6). Special tools are defined as those tools not listed in the Federal Supply Catalog (copies of this catalog may be consulted in the office of the Defense Contract Administration Services Management Area (DCASMA)). If coatings are used, such coatings shall be easily removable by means of a soldering gun or iron without damage to the printed wiring board. Coatings which emit toxic or harmful vapors or gases under heating conditions shall not be used (see 3.4.1).

3.6.14 Relays. Relays shall be in accordance with requirement 57 of MIL-STD-454, except as otherwise specified in MIL-STD-242. Those selected shall operate directly from the normal power supply of the equipment, within the tolerance characteristics thereof. In addition, relays shall preferably be of the sealed, plug-in type, firmly secured in the mounting sockets. Relays containing mercury in any form shall not be used. The devices shall operate satisfactorily under all conditions of usage, shall make firm, low-resistance contact, and shall retain their setting in any position of the equipment and under any service condition. Contacts shall be noncorrosive and, where practicable, of the self-cleaning type.

3.6.15 Synchros. Synchros shall be in accordance with MIL-S-20708 and as specified herein. Output synchros shall be selected in accordance with table I herein or in accordance with MIL-STD-710 and shall be as required for the intended application. MIL-HDBK-214 may be used for informational purposes in reference to synchros, whereas MIL-HDBK-225 shall be used for guidance in synchro applications and installation. The maximum permissible loads as specified in table I for various converter configurations shall be adhered to.

3.6.16 Servomotors. Servomotors shall be grease sealed ball bearing types. Selection of 60 and 400 Hz units shall be in accordance with MIL-S-22432.

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3.6.17 Prevention of SERVTORQ motoring. Motoring is a phenomenon of the SERVTORQ and synchro receiver continuing to run at a speed which may be up to 100 revolutions per minute (r/min) when the rotor is energized in the absence of a signal input to the stator.

3.6.17.1 Preventive measures. Unless otherwise specified (see 6.2.1), motoring shall be prevented. Means shall include, for example, a sensing device (such as a relay or solid state device) for removal of a rotor voltage (both R1 and R2 connections) in the absence of stator input (S1, S2 and S3 terminals). Relays, if selected, shall be in accordance with 3.6.14 and requirement 57 of MIL-STD-454.

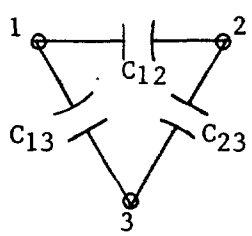
3.6.18 Synchro capacitors. When specified in the contract or order (see 6.2.1), the correct synchro capacitors shall be provided as part of the converter, delta connected across the stator leads (S1, S2 and S3) of differential synchros and control transformers. Each capacitor element shall be rated for 600 volts direct current (Vd.c.) for 60 Hz synchros and 1,000 Vd.c. for 400 Hz synchros. The capacitance value of each element shall be determined in accordance with DDS 430-2. Table II illustrates representative values of capacitors for specific synchro units.

TABLE II. Representative values of synchro capacitors for specific synchro units.

Synchro type	Capacitance per leg in microfarad (μF)
15CT6	0.34
18CT6	.24
23CT6	.34
11CT4	.036
15CT4	.022
18CT4	.014
23CT4	.014

NOTE: Capacitance per leg is defined as meaning the capacitor element in each leg of the delta may have the value indicated above for a specific synchro unit, upon completion of the calculations specified in DDS 430-2.

Example: type 15CT6, 0.34 μF per leg



$C_{12} = C_{13} = C_{23} = 0.34\ \mu F$

Where specified in table II, the synchro capacitors shall have the values noted within plus or minus 10 percent, as variations may occur between units of different origin. Each synchro capacitor shall have the capacitance of each of its three capacitor elements balanced within 1 percent, as measured by the

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divergence of the capacitance of any one leg from the arithmetical mean of the three. Capacitors shall be in accordance with MIL-C-39022, using MIL-HDBK-225 as guidance, and shall be mounted within the converter enclosure.

3.6.19 Synchro electrical zero and equipment mechanical zero. In order to insure that synchro components will work properly together in a system, it is essential that they be correctly connected and alined with respect to each other, and to the device or parent equipment with which they are used. Electrical zero is the reference point for the device or parent equipment using synchro components depending upon the particular application of the equipment involved. Unless otherwise specified by the contracting activity (see 6.2.1), whenever a synchro component is used, either as a transmitter or receiver, the synchro electrical zero and the equipment mechanical zero, shall be physically positioned to the same point. Synchros shall be electrically zeroed by the method as specified in MIL-HDBK-225 prior to delivery of equipment.

3.6.20 Synchro connections and markings. Synchro transmitters shall be connected to the converter terminals and the ship wiring, (where either may be marked with typical ship circuit designations, terminal "111" to ship wiring "111" and so forth) in such a manner that an increasing value (angular displacement) in the peripheral equipment synchro shaft (counterclockwise rotation) would cause a counterclockwise rotation in a standard synchro receiver connected to the ship wiring through the converter terminals (terminal R1 to ship wiring "110", R2 to "110(R)", S1 to "111", S2 to "112" and S3 to "113"). Direction of rotation shall be determined while facing the shaft end of the synchro. Synchro torque receivers and control transformers shall be connected to the converter terminals in such a manner that, with direct connection between the converter terminals and the ship wiring, counterclockwise rotation of a standard transmitter having direct connection to the ship wire (R1 to "110", R2 to "110(R)", S1 to "111", S2 to "112" and S3 to "113") shall cause counterclockwise rotation of the converter dial or pointer. Conversely, for a synchro that requires clockwise rotation for increasing signal, synchro terminals S1, S2 and S3 shall be connected to the converter terminals corresponding to ship wire numbers 113, 112 and 111, respectively. (See figure 10 and MIL-HDBK-225 for additional connection data.)

3.6.21 Electronic devices and components.

3.6.21.1 General requirements.

3.6.21.1.1 Breakdown voltage ratings. For transistors, the type selected shall have a collector-to-emitter and collector-to-base breakdown voltage rating at least three times the voltage in the circuit of use. For diodes and similar devices (for example, thristors), except zener diodes, the device chosen shall have a peak inverse or peak reverse voltage rating not less than four times the voltage of the circuit wherein used. Microcircuits (also referred to as integrated circuits and integrated devices) selected for application shall have a supply voltage rating not less than 1.5 times the voltage in the circuit of use. Voltage ratings of devices shall be increased upward as required, in addition to the use of circuit protection, in the event transient or spike voltages are present in a network (see 3.6.21.1.3).

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3.6.21.1.2 Current ratings. Current ratings of semiconductor devices (discrete and integrated) selected for application shall be not less than twice those encountered in the circuit wherein used. Should surge currents be present in a network, the applicable ratings of all devices shall be increased upward as required, with particular attention paid to various surge current ratings expressed by device manufacturers. Surge current ratings of devices shall be not less than twice those encountered in the circuits wherein used. Current limiting means shall be employed whenever necessary to preclude damage to solid state components, especially if transients occur in the network (see 3.6.21.1.3).

3.6.21.1.3 Transient protection. Should transient or spike voltages and surge currents be present in a network, the applicable rating, of solid state devices shall be increased upward as required. Transistors and microcircuits shall also be protected against transients by suitable means, for example, back-biased diodes, to avoid exceeding device breakdown voltages. With respect to transistors and integrated devices, when converters are tested in accordance with 4.6.1.9, no damage to the equipment shall occur and dynamic accuracy shall be in accordance with 3.7.1. Circuit analysis shall also consider avoidance of exceeding device power dissipation under transient conditions (see 3.6.21.1.4).

3.6.21.1.4 Power dissipation. In audio applications, the instantaneous power required to obtain the desired output power shall in no case exceed one-half of the rated maximum power dissipation of a discrete device. For switching application, the instantaneous power may not exceed one-half of the rated maximum power dissipation of a discrete device only after and to the extent that an analysis will show that the peak instantaneous power and the repetition rate of this peak results in an average power one-half or less of the rated maximum power dissipation of the discrete device. For microcircuits, the instantaneous power specified above for audio and switching applications shall not exceed one-half the maximum rated power dissipation of the complete microcircuit. For voltage dropping devices the power dissipation shall be at least 50 percent greater than the maximum design value.

3.6.21.1.5 Temperature considerations. The effect of the ambient temperature specified below on solid state device ratings and equipment performance shall be determined and considered in the design. The maximum rated function temperature of electronic devices shall not be exceeded under any condition of use. The requirements specified in 3.6.21.1.1 through 3.6.21.1.4, 3.6.21.1.6, and 3.6.21.1.7 shall be considered in the equipment design. The equipment is intended for use in an ambient temperature range of 40 to 149°F.

3.6.21.1.6 Mounting. Power devices shall be mounted to allow maximum heat transfer. The desired mounting is directly on its heat sink or on the metal of a chassis. A thermal conductive compound shall be applied between the mating surfaces. Equipment design shall be investigated to determine if other components require such mounting. Parts provided with pigtail leads (for example, various types of diodes, transistors, and integrated circuits, and small resistors and capacitors), other than those mounted on printed wiring boards, shall be secured between solder-type terminals on part mounting boards in accordance with the following. The arrangement of these parts shall be such as to limit the number of leads terminating at any one terminal to three, except for double turret terminals which may terminate three wires per section (turret). The attachment and number of leads shall be in accordance with

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requirement 5 of MIL-STD-454. The clearance between the soldered connection and the body of the part shall be not less than 3/16 inch. The basic requirements shall predicate that parts with pigtail leads shall not be mounted by means of the wire leads unless other mechanical support is provided, except that parts weighing less than 1/2 ounce may be secured by only their leads. The leads shall be as short as practicable, keeping in mind the clearance limitation of 3/16 inch noted above, and in no case shall the leads be longer than 1 inch. Parts having a diameter larger than 1/2 inch shall be securely fastened by one or more clamps. Kinks or other slacks shall be provided in the mounting leads to permit thermal expansion of the terminal or connection board without stressing the parts. Parts shall be arranged to permit the replacement of any component without removal of or damage to adjacent parts (see 3.6.1).

3.6.21.1.7 Thermal considerations. Current shipboard considerations shall require an integrated or self-contained thermal design for electronic assemblies, in order to properly dispose of heat generated therein. Portions of the heat removal path shall be designed with as low a resistance to the flow of heat as possible. Sufficient heat sink shall be provided such that the performance of the equipment shall be as specified at the maximum specified ambient temperature (see 3.6.21.1.5 and 3.6.21.1.6). Convection cooling shall be a prime consideration for equipment design. Circuit boards shall be mounted vertically to allow maximum transfer to the circulating air.

3.6.21.2 Discrete semiconductor devices. Discrete semiconductor devices, such as transistors and diodes, shall be in accordance with MIL-STD-701 with respect to choice and application and shall be in accordance with MIL-S-19500. Only silicon devices shall be used. The selection of same shall include the requirements specified in 3.6.21.1 in the design analysis.

3.6.21.3 Microcircuits. Microcircuits or integrated circuit devices shall be selected in accordance with requirement 64 of MIL-STD-454. The selection of same shall include the requirements specified in 3.6.21.1 in the design analysis.

3.6.22 Converter signal input impedance. The secondary load on the converter input control transformers shall be greater than 10,000 ohms for 60 Hz units and greater than 15,000 ohms for 400 Hz units to prevent excessive loading of the converter signal source transmitters.

3.7 Performance.

3.7.1 Accuracy. The accuracy of the synchro converters shall be such that when tested as specified herein, the errors shall not exceed those specified in the notes of the applicable figures (see figures 1 through 7). For multiple input speeds, "input speed" shall be the highest speed on any one channel. For example, a converter having 1 and 36 speed inputs and 1 and 36 speed outputs, not isolated from each other, shall be considered to have a 36 speed input with a 1 speed output accurate to 0.5 degree and a 36 speed output (1 to 1) accurate to 1 degree (see figure 6). Another converter having isolated inputs of 1 and 36 speed, and outputs of 1 and 36 speed, shall be considered to have two separate 1 to 1 channels, each output being accurate to 0.5 degree (2 channel converter in accordance with figure 1) (see 1.2.1).

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3.7.2 Synchronization. Converters shall be self-synchronous from all displacements. When tested in accordance with 4.6.1.5, the output synchros shall synchronize with the input synchros, stabilize, and attain the static accuracy specified in 3.7.1. Single- and double-speed converters and differential converters shall attain synchronism, as specified herein, in not more than 7 seconds. Triple-speed converters shall be self-synchronous; however, a time limit for achieving synchronism shall not apply.

3.7.3 Phase shift. When tested in accordance with 4.6.1.6, the converters shall operate in accordance with 3.7.1 for all phase shifts between 15 degrees lag and 5 degrees lead caused by external synchro units in the synchro transmission system. For the purpose of this requirement, phase shift is defined as the electrical angular displacement of the fundamental component of the primary (stator) voltage from the secondary (rotor) voltage applied to the input synchros as specified in MIL-S-20708.

3.7.4 Effect of synchro bus load on synchro control transformers. In order to prevent excessive reflective errors on the synchro bus, the secondary load on the synchro control transformers shall be greater than 10,000 ohms for 60 Hz units and greater than 15,000 ohms for 400 Hz units.

3.7.5 Effect of magnetic fields. When tested in accordance with 4.6.1.7, converters shall operate within the accuracy limits specified in 3.7.1 while located in an ambient magnetic field of 5 gauss.

3.7.6 Power supply tolerance. When tested in accordance with 4.6.1.8, converter shall be in accordance with the static and dynamic accuracy requirements of 3.7.1.

3.7.7 Transients. When tested in accordance with 4.6.1.9, no damage shall occur to the converters and dynamic accuracy shall be in accordance with 3.7.1.

3.7.8 Electromagnetic interference. When tested in accordance with MIL-STD-462, converters shall be in accordance with the requirements of MIL-STD-461.

3.7.9 Airborne noise. When tested in accordance with 4.6.3, converters shall be in accordance with the requirements of MIL-STD-740 for grade C equipment.

3.7.10 Structureborne noise. When tested in accordance with 4.6.4, converters shall be in accordance with the requirements of MIL-STD-740 for type 3 equipment.

3.7.11 Accelerated life. The static and dynamic accuracy of the converter shall be in accordance with 3.7.1 during each of the operational modes specified in 4.6.5.

3.7.12 Vibration. When tested in accordance with 4.6.6, no damage shall occur and accuracy shall be in accordance with 3.7.1.

3.7.13 Shock. No damage shall occur during the shock test specified in 4.6.7. During the test, dynamic accuracy shall be in accordance with 3.7.1 and following the test, static accuracy shall be in accordance with 3.7.1.

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3.7.14 Dielectric withstanding voltage. When tested in accordance with 4.6.8, the converter shall withstand the test potentials between application points (specified). The converter shall display no evidence of insulation breakdown or of internal arcing nor shall winding leakage current exceed 1 milliampere peak. The leakage current limit shall not include the current drawn by the test equipment capacitance. Immediately after meeting this requirement the converter shall meet the insulation resistance requirement.

3.7.15 Insulation resistance. When tested in accordance with 4.6.9, insulation resistance of electrical circuits shall be not less than 10 megohms.

3.7.16 Synchro connections. Connections and markings specified in 3.6.20 shall be determined by the test specified in 4.6.10.

3.7.17 Input operation. The design specified in 3.6.18 shall be determined in accordance with 4.6.11.

3.7.18 Temperature and humidity. The tests specified in 4.6.12 shall be conducted to determine compliance with 3.7.1.

3.8 Designation and marking. Identification plates, information plates, and other designation plates and markings for units, subassemblies, and parts shall be in accordance with requirement 67 of MIL-STD-454, except that reversed etched plates shall not be used. Serial numbers on plates for major parts or subassemblies (for example, synchro, including SERVTORQ units) shall be required and shall be assigned sequentially. Plates shall include the manufacturer's code, electrical parameters, item name, type designation, National stock number, and so forth.

3.9 Technical data. The contractor shall prepare drawings (see 3.9.1), technical manuals (see 3.9.2) and reliability predictions (see 3.9.1) in accordance with the data ordering documents included in the contract or order (see 6.2.2).

3.9.1 Drawings. Drawings shall cover the mechanical features of the converter and provide the following information:

(a) Permit the contracting activity to determine that the design complies with the requirements of the specifications regarding materials and parts used, size, method of fabrication, mounting and mating of assemblies and subassemblies, suitability for shipboard installation, maintenance, and modification.
(b) Provide the installing activity with clear and accurate information regarding location and sizes of mounting bolt holes, location of connector for connection of switchboard internal harness wiring, clearances, and access requirements.
(c) Provide the in-service engineering agent and supply activities with detail information concerning the various parts used in the converter, so that replacement parts may be acquired or additional converters may be fabricated, when required.

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3.9.1.1 The mechanical assembly drawing shall include dimensioned outline front view, plan view, end views, and such other views as are required to show clearly the details of mechanical design, construction, and assembly of one complete converter. Overall dimensions and mounting bolt holes shall be as shown on figure 8.

3.9.1.2 Part identification. Each replaceable part shall be identified with the following minimum information on the assembly drawing:

(a) Nomenclature or description.
(b) Part or identifying number as used by the actual manufacturer of the part (not the converter manufacturer's part number).
(c) Actual specification name or code identification number.
(d) Government specification number (or commercial where there is no applicable Government specification).
(e) Material (unless several materials are involved).
(f) Mechanical, electrical and physical characteristics.

The above information may be tabulated in a parts list or shown on the detail drawing of the part. Where several assemblies requiring varying numbers of the same parts are shown on a drawing, each part shall be listed only once in the parts list, each assembly shall be assigned a number and the "No. Req." column shall be replaced by a "No. Req. for No. __" column for each assembly. The weight of the converter shall be shown.

3.9.1.3 Gearing and electrical schematic diagrams shall be shown on the assembly drawing.

3.9.2 Technical manuals. Contractor shall prepare technical manuals if required (see 6.2.1). MIL-HDBK-225 shall be referenced for synchro maintenance information. When specified (see 6.2.1 and 6.2.2), loose leaf sheets shall be provided as complete sets in the same format as for the manuals to completely cover maintenance and operation of the converter.

3.10 Workmanship. Workmanship shall be in accordance with requirement 9 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality program. The contractor shall provide and maintain a quality program acceptable to the Government for supplies and service covered by this specification. The quality program shall be in accordance with MIL-Q-9858.

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4.1.2 Reliability. Reliability principles and techniques shall be implemented in accordance with requirement 35 of MIL-STD-454.

4.1.2.1 A reliability prediction shall be made on the equipment based upon electrical/mechanical/thermal stress analysis to assess the mean-time-between failure (MTBF) (see 6.2.2). The prediction shall be performed at the maximum ambient temperature condition and for the Naval sheltered environment. The prediction shall be in accordance with MIL-HDBK-217, and shall include the part reference designation, part type, part nomenclature, applicable Military part specification, part electrical and mechanical rating, design electrical and mechanical stress, electrical and mechanical stress ratio, base failure rate, failure rate application factors, final failure rate, and any other information used to determine part failure rates. Where MIL-HDBK-217 does not apply, other data sources may be used subject to review by the contracting activity. When such alternate sources of data are proposed, sufficient information shall be provided to enable the contracting activity to evaluate its validity and applicability.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Qualification inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Qualification inspection. Qualification inspection shall be conducted at a laboratory satisfactory to NAVSEA. Qualification inspection shall consist of the examination and tests specified in table III. Tests shall be conducted in the order specified in table III. Engineering analysis will be conducted by a NAVSEA representative and will be acccomplished prior to qualification inspection.

TABLE III. Qualification inspection.

Inspection	Requirement paragraph	Test paragraph
General examination	3.4, 3.5, 3.6 and 3.7	4.5
Accuracy and oscillations 1/	3.7.1	4.6.1 through 4.6.1.4
Synchronization 1/	3.7.2	4.6.1.5
Phase shift	3.7.3	4.6.1.6
Effect of synchro bus load	3.7.4	4.6.1.2
Effect of magnetic fields	3.7.5	4.6.1.7
Power supply tolerance 1/	3.7.6	4.6.1.8
Transients	3.7.7	4.6.1.9
Electromagnetic interference 1/	3.7.8	4.6.2
Airborne noise	3.7.9	4.6.3
Structureborne noise	3.7.10	4.6.4
Accelerated life	3.7.11	4.6.5
Vibration	3.7.12	4.6.6
Shock	3.7.13	4.6.7
Dielectric withstanding voltage	3.7.14	4.6.8

See footnote at end of table.

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TABLE III. Qualification inspection. - Continued

Inspection	Requirement paragraph	Test paragraph
Insulation resistance 1/	3.7.15	4.6.9
Synchro connections	3.7.16	4.6.10
Input operation	3.7.17	4.6.11
Temperature and humidity	3.7.18	4.6.12

1/ See 4.3.1.1(c).

4.3.1 Samples for qualification inspection. For qualification inspection of individual types, one converter of that type shall be submitted for qualification inspection.

4.3.1.1 Qualification of one type of converter will be considered for qualification of other types of converters provided the following conditions are met:

(a) The enclosure is the same for all types of converters.
(b) The electronics module, drive motor, and switching network used for all type units is interchangeable or a single integral switch is used to convert from one type to the other.
(c) The unit passes all tests for one type operation and after conversion, (see (b)) above, passes the tests indicated with a footnote for the other type operation (see table III for qualification inspection and table IV for quality conformance inspection).

4.3.1.2 Qualification engineering analysis. The qualification applicant shall develop engineering data and drawings of sufficient detail to allow NAVSEA to determine that the design requirements specified in 3.4, 3.5 and 3.6 have been met. The engineering data shall include a complete parts and materials list, identifying the manufacturer's approved sources. Standard parts or materials (see 3.3.2) shall be identified by military part number and qualified products list number as applicable. Specifications, drawings or other engineering data to fully identify the parts or materials shall be developed for nonstandard items. These data, the reliability data (see 4.1.2), and the inspection data (see 4.3.1.1) shall be the basis for determining the suitability for inclusion of the applicant's part number on the qualified products list.

4.4 Quality conformance inspection. Converters on the contract or order shall be subjected to group A examination and tests specified in table IV. Group B inspection will be required by NAVSEA when the basic design of the equipment or when a vital part has been changed from that of the unit qualified. For example, a change from one design of amplifier unit to another design using different components shall be regarded as a basic design change. Another example, a change in gearing from 1- and 36-speed to 1- and 5-speed or from a 1- and 36-speed input and output to 1- and 36-speed input and 2- and 36-speed output shall not be regarded as a basic design change. One complete converter shall be subjected to each of the group B tests as specified in table IV. Any converter selected for group B tests shall not be used as a production converter.

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TABLE IV. Quality conformance inspection.

Inspection	Requirement paragraph	Test paragraph
Group A		
General examination	3.4, 3.5, 3.6 and 3.7	4.5
Accuracy and oscillations	3.7.1	4.6.1 through 4.6.1.4
Synchro connections	3.7.16	4.6.10
Input operation	3.7.17	4.6.11
Synchronization	3.7.2	4.6.1.5
Phase shift	3.7.3	4.6.1.6
Effect of synchro bus load	3.7.4	4.6.1.2
Group B		
General examination	3.4, 3.5, 3.6 and 3.7	4.5
Power supply tolerance 1/	3.7.6	4.6.1.8
Electromagnetic interference	3.7.8	4.6.2
Airborne noise	3.7.9	4.6.3
Structureborne noise	3.7.10	4.6.4

1/ See 4.3.1.1(c).

4.4.1 Rejection. The results of each test shall be compared with specification requirements. Failure to conform to this specification for any group A or B test shall be counted as a defect, and the converter shall not be offered for delivery.

4.5 General examination. The completed unit shall be given a thorough examination to determine that it conforms to the applicable specifications and drawings with respect to material, finish, workmanship, construction, assembly, dimensions, weight, and marking of identification and information plates. The examination shall be limited to those examinations that may be performed without dissembling the unit in such a manner that its performance, durability, or appearance would be affected. The examination shall include a check of operating controls, circuit functions, synchro connections in accordance with 3.6.20 and test facilities. The examination shall determine that the equipment is in accordance with the requirements of 3.4, 3.5, 3.6 and 3.7. The contractor shall develop for the DCASMA a detailed written description of the materials, design and manufacturing processes of the unit(s) to be examined and tested. Samples submitted shall be representative of the manufacturer's normal production. The contractor shall establish adequate calibration of test equipment to the satisfaction of the DCASMA.

4.6 Tests.

4.6.1 Accuracy and oscillations. The purpose of these procedures and tests is to determine compliance with 3.7.1. MIL-S-20708 may be referred to for other test setups.

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4.6.1.1 Accuracy. A suggested circuit for accuracy testing of a SERVTORQ is shown on figure 11. With the synchro bridges set on zero, the reference (1) unit shall be mounted in a synchro index stand which can position the SERVTORQ rotor within plus or minus 15 seconds of arc. The fine adjustment control shall be adjusted until a minimum null reading is indicated on phase sensitive voltmeter (PSVM) (1). The dial indicator of the index stand shall then be mechanically positioned to the zero heading. After the indicator has been set on zero, the referenced (1) unit shall be repositioned by adjusting the fine adjustment control of the index stand until a minimum reading is indicated on PSVM (2). The reading which appears on the dial indicator of the index stand shall be the electrical error. The index stand and two synchro bridges shall then be adjusted to the next 5 degrees heading. If the two reference units of figure 11 do not have an electrical error of more than 2 minutes of arc each, their respective errors shall be taken into account and the SERVTORQ error corrected accordingly. It should be noted that reference (1), bridge (1), and PSVM (1) may be collectively replaced by a synchro standard.

4.6.1.2 Synchro loading. Qualification tests for overall static and dynamic accuracy shall be determined with no load on the output transmitters and with specified loads on the same. The specified loads on each output transmitter shall consist of two synchro receivers of the same size type and frequency as the transmitter. Quality conformance tests on production units (groups A and B) shall be made with no loads on the output transmitters.

4.6.1.3 Overall static accuracy. This test, not applicable to differential converters, shall be determined by turning the test director transmitters or synchro standard controls slowly in not greater than 10 degree of arc steps, referred to the low-speed transmitter or as applicable in the standard, and measuring the error in accordance with 4.6.1. The errors at six equally spaced steps per revolution, in both directions, shall be recorded.

4.6.1.4 Overall dynamic accuracy. This test, not applicable to differential converters, shall be conducted by operating test director or standard under the following conditions and measuring the error in accordance with 4.6.1. Constant velocity inputs refer to the lowest speed transmitter in the director. Amplitude in simple harmonic motion is defined as the maximum excursion in each direction about the central heading.

(a) Constant velocity of 32 degrees per second for converters having single- or double-speed output.
(b) Simple harmonic motion of 40.0 degree amplitude and 8.0 second period for converters having single- or double-speed output.
(c) Constant velocity of 6000 yards per second for 1, 18, and 648-speed equipment.
(d) Simple harmonic motion of 40.0 degree amplitude and 25 second period for 1, 18 and 648-speed equipment.

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4.6.1.5 Synchronization. The amplifier unit of the converter shall be in a ready condition. Power to the servomotor in the converter shall be turned off while the transmitters in the test director are set 180 degrees out of correspondence with the input synchros of the converter. Simultaneously, power shall be restored to the servomotor and a timing device started. The total time for the converter to synchronize with the test director (see 3.7.2) shall be measured when testing converters having single-speed or double-speed output. The performance of three-speed output converters shall be observed to determine that synchronization occurs, but need not be timed.

4.6.1.6 Phase shift. Synchro transmitters in the test director shall be loaded to produce the phase shifts specified in 3.7.3 and shall be connected to the converter as specified in 4.6.1. The test director shall be positioned and turned as specified in 4.6.1.3 and 4.6.1.4. The converter output error shall be determined in accordance with 4.6.1.

4.6.1.7 Effect of magnetic fields. Static and dynamic accuracy tests (see 4.6.1, 4.6.1.3, and 4.6.1.4) shall be performed with the converter located in the specified magnetic field (see 3.7.5). No synchro loads shall be applied to the converter during this test.

4.6.1.8 Power supply tolerance. Converter shall be operated in accordance with 4.6.1.4(b) or 4.6.1.4(d) as applicable, while energized from a type I power source specified in DOD-STD-1399, section 300, in which the voltage and frequency are at the upper tolerance limit (108 percent voltage and 105 percent frequency). The test shall be repeated with the voltage and frequency at the lower tolerance limit (92 percent voltage and 95 percent frequency).

4.6.1.9 Transients. With the converter operating in accordance with 4.6.1.4(b) or 4.6.1.4(d), as applicable, a 2500 V pulse, as specified in DOD-STD-1399, section 300, shall be superimposed on the supply voltage at the converter terminals. Following the pulse discharge, synchro dynamic errors shall be observed and the converter shall be examined in accordance with 3.7.7.

4.6.2 Electromagnetic interference. Converter shall be tested in accordance with the requirements specified in 3.7.8.

4.6.3 Airborne noise. Converter shall be tested in accordance with the requirements specified in 3.7.9.

4.6.4 Structureborne noise. Converter shall be tested in accordance with the requirements specified in 3.7.10.

4.6.5 Accelerated life test. Accelerated life test shall be conducted for a period of 250 hours. Minimum period of operation shall be 7 hours except as indicated. Conditions of operation shall be varied as specified in 4.6.1.4(b) or 4.6.1.4(d), as applicable. The 250 hour test shall be divided as shown in table V. Before starting the tests, an accuracy test shall be conducted and recorded.

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TABLE V. Accelerated life test.

Hours 1/	Ambient temperature during test	Test conditions	Angle of tilt
0-25	75°F nominal (room temperature)	Constantly varying	Normal
After 25	Vibration test (see 4.6.6), accuracy check test, and short test runs (1/2 hour each) at undervoltage and under-frequency and at overvoltage and over-frequency (see 4.6.1.9)		
25-85 continuous	40°F	Constantly varying	Normal
85-150 continuous	149°F at 95 ± 4 percent relative humidity and then chamber shut down for alternate 1-hour periods. Hot spots, internal air and ambient temperature shall be continuously scanned.	Constantly varying	Normal
150-175	75°F nominal (room temperature)	Constantly varying	Incline 60 degrees forward from vertical
175-200	75°F nominal (room temperature)	Constantly varying	Incline 60 degrees back from vertical
200-225	75°F nominal (room temperature)	Constantly varying	Incline 60 degrees right from vertical
225-250	75°F nominal (room temperature)	Constantly varying	Incline 60 degrees left from vertical

1/ Time periods may be varied to suit laboratory test schedule.

Upon completion of above tests, an accuracy test shall be conducted and recorded. Immediately after the above endurance test is completed the insulation resistance test, shock test and accuracy check test shall be conducted in the order specified.

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4.6.6 Vibration. Converter shall be subjected to type I vibration as specified in MIL-STD-167-1, except that the exploratory vibration test shall be omitted. During the vibration, the converter shall be operated in accordance with 4.6.1.4(b) or 4.6.1.4(d), as applicable. Subsequent to the test, the converter shall be examined for damage and the static and dynamic accuracy shall be determined in accordance with 4.6.1.3 and 4.6.1.4.

4.6.7 Shock. Converter shall be tested in accordance with MIL-S-901 for grade A, class I, type B equipment. Converter shall be operated as specified in 4.6.1.4(b) or 4.6.1.4(d), as applicable. Following each blow, the converter shall be examined for damage and the dynamic errors shall be observed. Following the test, the static accuracy shall be determined in accordance with 4.6.1.3 to ensure compliance with 3.7.13.

4.6.8 Dielectric withstanding voltage. Converters shall be tested in accordance with method 301 of MIL-STD-202 and the requirements of the applicable test potential shall be applied between the points designated. The test potential shall be raised slowly (minimum time 3 seconds) from zero to the specified value. The peak value of the test potential throughout this test shall not exceed 1.5 times the specified root mean square (rms) test potential. The test equipment employed shall be such as to differentiate between winding leaking current and surge discharge current. Following this test, the insulation resistance test shall be performed in accordance with 4.6.9.

4.6.9 Insulation resistance. Insulation resistance shall be measured by means of a device that applies a voltage (in the range of 1.5 to 50) compatible with the circuit being tested while indicating insulation resistance. This measurement shall be made between live circuits and ground. Insulation resistance shall be in accordance with 3.7.15.

4.6.10 Synchro connections. Converter shall be examined carefully, by disassembly as necessary, in accordance with 3.7.16 and for comparison with the drawings for the equipment.

4.6.11 Input operation. Converter shall be carefully examined in accordance with 3.7.17.

4.6.12 Temperature and humidity. The following test conditions shall apply:

(a) Low temperature test shall be conducted with the unit in a chamber for 4 hours at 40°F. Upon reaching the 4-hour point, the converter shall be energized and accuracy determined in accordance with 4.6.1.3 in order to ensure compliance with 3.7.18.

(b) High temperature test shall be conducted with the converter, panel-mounted and operating with a constantly varying input synchro signal of simply harmonic motion of 40 degrees amplitude and 8 second period (see 4.6.1.4). In addition, the heat dissipated from all six sides of the converter shall be restricted by means of low heat conducting bulkheads placed adjacent to the sides of the converter. Converter shall be operated uninterruptedly for 24 hours at no less than 149°F at a relative humidity of at least 95 percent. Hot spot

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temperatures shall not exceed 221°F, which dictates that the temperature rise shall not exceed 72°F near the end of the test. The number and location of hot spot sensors shall be determined by scanning the various components in the unit to determine maximum hot spot temperatures. Not less than three hot spot sensors shall be attached to heat generating components in the converter. At anytime during and after the test, the converter shall meet the static and dynamic tests in accordance with 4.6.1.3 and 4.6.1.4.

4.6.12.1 Internal examination. The converter and the internal mechanism shall be examined in accordance with 3.10 and those requirements for 3.4, 3.5 and 3.6 that are amenable to visual inspection without disassembly of the mechanism. Any deterioration or damage that could in any manner prevent the converter from meeting functional operation or maintenance requirements during service life shall be cause for rejection.

4.7 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.4.)

5.1 Preservation-packaging, packing, and marking. Equipment and accessories, repair parts and publications shall be preserved-packaged by levels A or C; packed by levels A or B, as specified (see 6.2.1), and marked in accordance with MIL-E-17555.

6. NOTES

6.1 Intended use. Converter is used for one or more of the following applications:

(a) To permit the operation of increased synchro loads.
(b) To permit the operation of a 400-Hz synchro load from a 60-Hz synchro bus or vice versa.
(c) To permit the transmission of a modified synchro signal.
(d) To permit the operation of a synchro load independent of the input synchro excitation.
(e) To minimize reflected errors on the input synchro signal bus.
(f) To provide a smooth and accurate transmission of synchro signals.

In addition, the attention of contractors is called to the fact that the equipment covered by this specification is intended for rigorous Naval shipboard service, where it is expected to withstand continuous use for long periods, under Military service conditions, without benefit of overhaul. The equipment is in each case a vital unit intended for extremely important usage by the Naval forces concerned. Failure at a critical moment results in a serious reduction

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of the battle efficiency of the ship involved. Emergency repairs afloat, or at distant bases, can seldom be made with sufficient celerity to avoid having the equipment out of service during a critical shipboard period or evolution.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type designation required (see 1.2 and 3.2.1).
(c) Fuse installation (see 3.6.5.1).
(d) Fuseholder installation (see 3.6.5.2).
(e) Energization of blown fuse indicator circuit (see 3.6.5.3).
(f) External electrical connections, specify connections as required (terminal board or connector) (see 3.6.6).
(g) If motoring need not be provided (see 3.6.17.1).
(h) If synchro capacitors are to be provided as part of converter (see 3.6.18).
(i) Synchro electrical zero and equipment mechanical zero are not positioned to the same point (see 3.6.19).
(j) 1/18X accuracy to be 0.2 degree or 0.5 degree (figures 3, 4, 5 and 7).
(k) Technical manual or quantity of looseleaf sheet required (see 3.9.2).
(l) Levels of preservation-packaging and packing required (see 5.1).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of FAR 52.227-7031 are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
3.9.1	Drawings, engineering and associated lists	DI-E-7031	Level 3 Design activity designation - contractor Drawing number - contractor Delivery of hard copies - contracting activity
3.9.2	Manual, technical, functionally oriented maintenance manual (FOMM)	DI-M-2045	MIL-M-24100
4.1.2.1	Reliability prediction	DI-R-7095	----

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(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in Qualified Products List QPL-24105 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362 and information pertaining to qualification of products may be obtained from that activity. Application for qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3.1).

6.3.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.

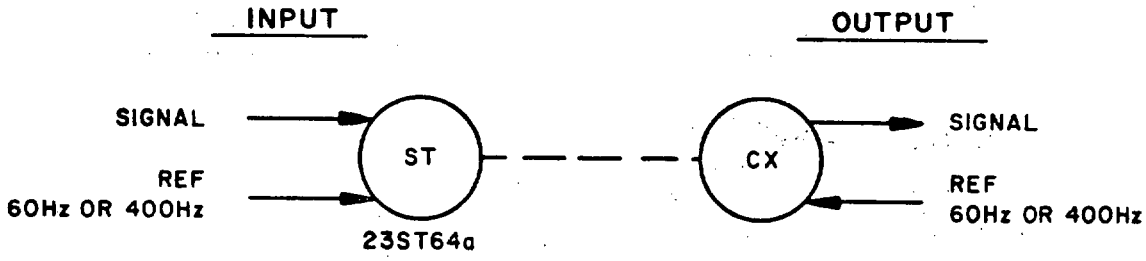
6.4 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
Navy - SH
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SH 11379

NOTES:

1. Output synchro may be CX, CDX, TX, TDX, or TRX of sizes 23, 31, or 37.

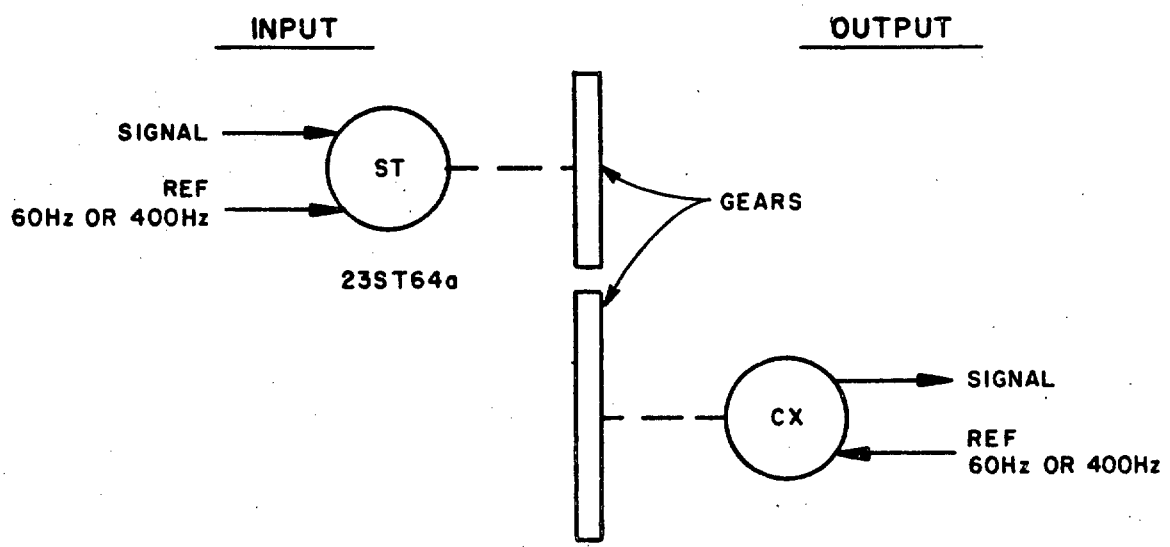
2.

If input is:	Output is:	Accuracy is:
1X	1X	0.5 degree
2X	2X	0.5 degree
36X	36X	0.5 degree
2K Yds	2K Yds	3 Yds
72K Yds	72K Yds	100 Yds
1296K Yds	1296K Yds	1800 Yds

3. Up to 3 independent channels may be provided in a one panel converter (see figure 8), provided output synchros are size 23, and not more than two of them are differentials.

FIGURE 1. Single input-single output, one, two, or three independent or isolated channels, direct coupled types.

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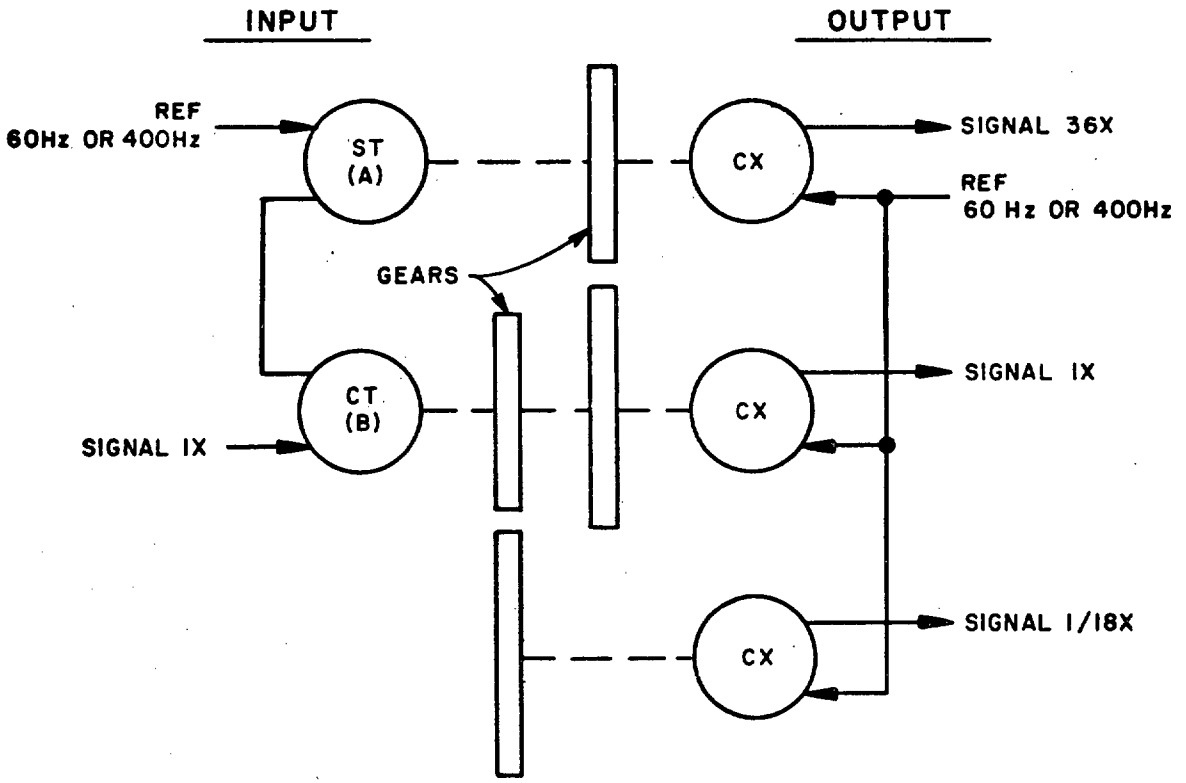
SH 11380

NOTES:

1. Output synchros may be CX, CDX, TX, TDX, or TRX of sizes 23 only.
2. Two channels may be provided in a one panel converter (see figure 8).
3. Accuracy is 1.0 degree.

FIGURE 2. Single input-single output, one or two independent or isolated channels, 2 to 1 or 1 to 2 gear ratio types.

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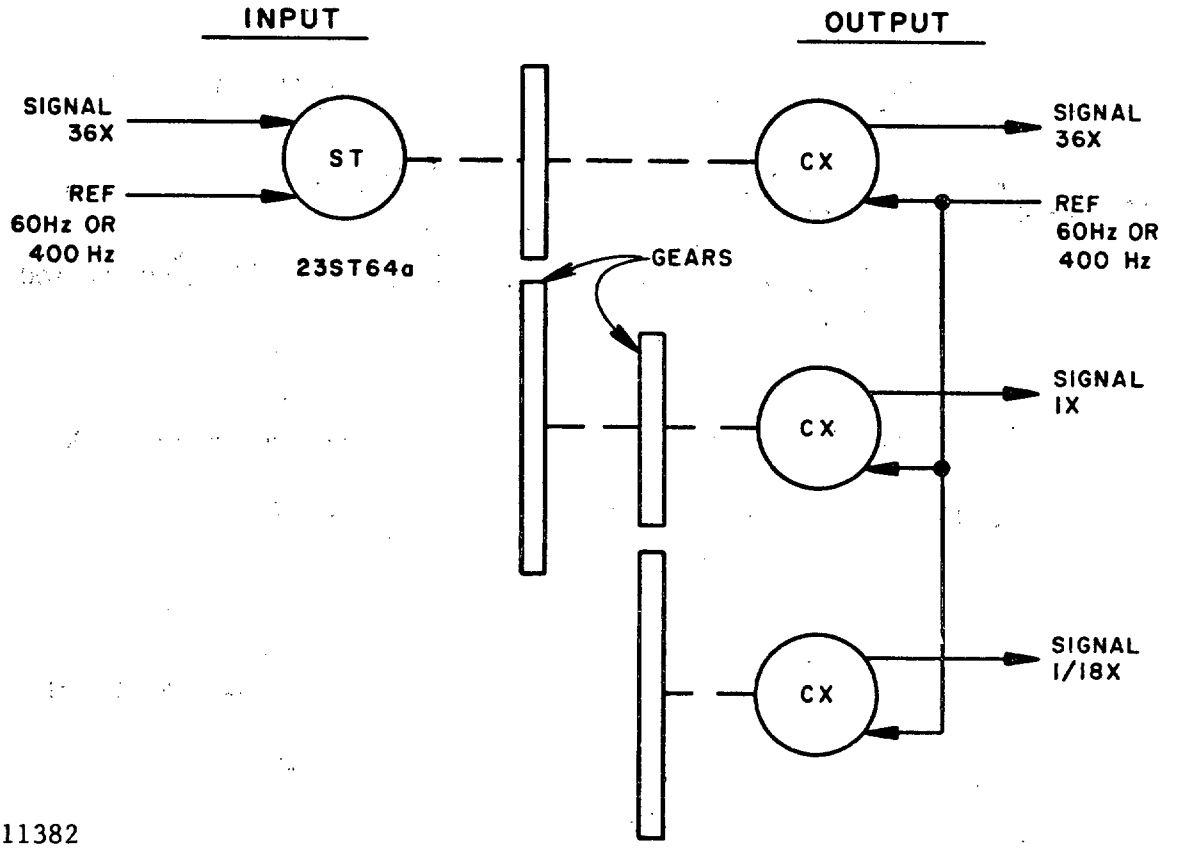
SH 11381

NOTES:

1. ST(A) and CT(B) are split version of standard 23ST64a.
   (A) contains d.c. torque motor and electronics.
   (B) is a CT only.
2. Output synchros may be CX, TX, or TRX, size 23.
3. Direct-drive no-speed switching network is required.
4. Converter size is for one panel (see figure 8).
5. 36X equivalent to 648X or 2K Yds/Rev.
   1X equivalent to 18X or 72K Yds/Rev.
   1/18X equivalent to 1X or 1296K Yds/Rev.
6. 36X accuracy = 36 degree or 200 Yds.
   1X accuracy = 1 degree or 200 Yds.
   1/18X accuracy = 0.2 degree or 720 Yds with anti-backlash gears, or 0.5 degree or 1800 Yds without anti-backlash gears (see 6.2.1).

FIGURE 3. Single input-triple output, nonisolated channel, 1 to 1/18, 1 and 36 gear ratio type.

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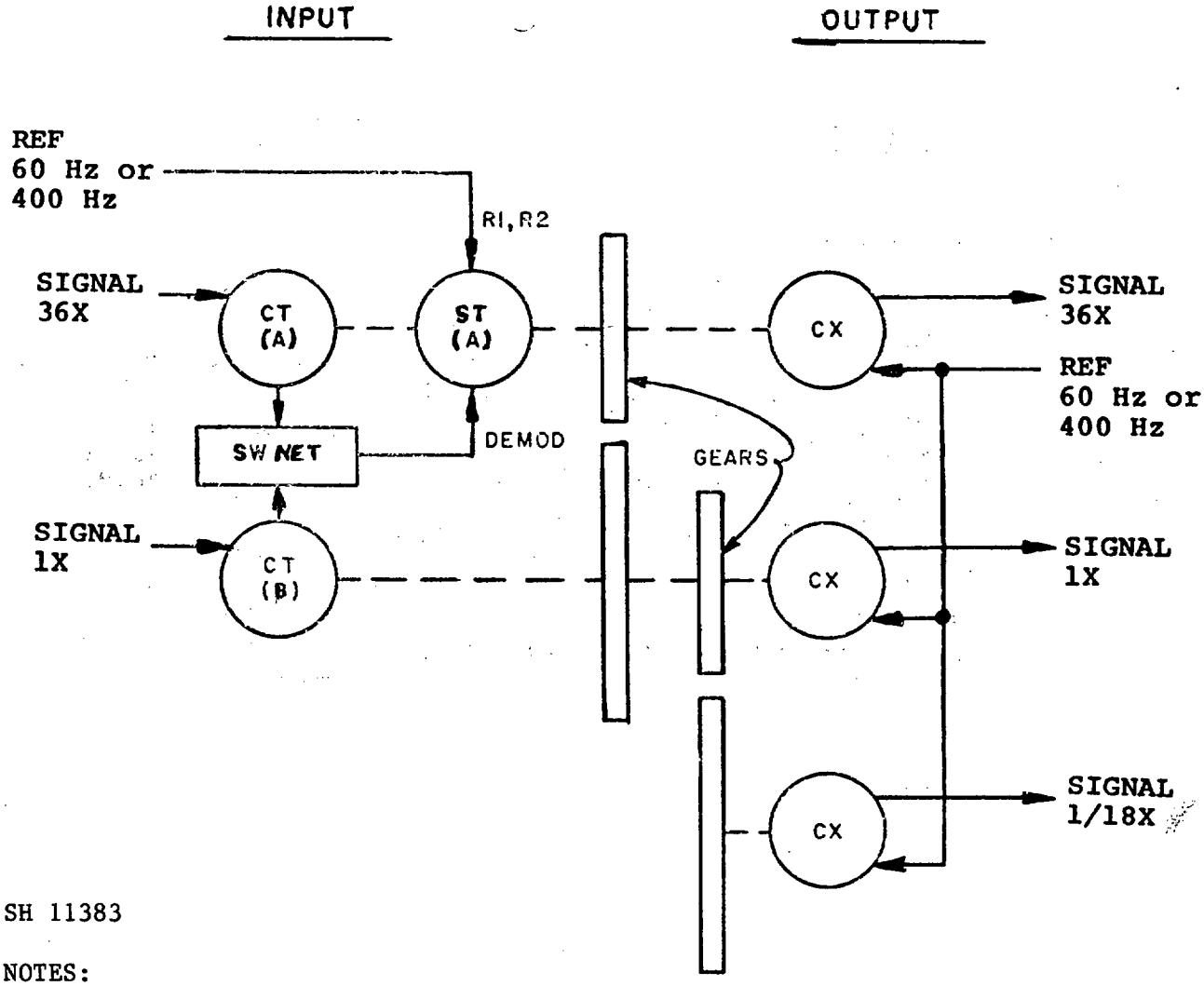
SH 11382

NOTES:

1. Output synchros may be CX, TX, or TRX, size 23.
2. Direct-drive no-speed switching network is required.
3. Converter size is for one panel (see figure 8).
4. 36X equivalent to 648X or 2K Yds/Rev.
   1X equivalent to 18X or 72K Yds/Rev.
   1/18X equivalent to 1X or 1296K Yds/Rev.
5. 36X accuracy = 1 degree or 6 Yds.
   1X accuracy = 0.5 degree or 100 Yds.
   1/18X accuracy = 0.2 degree or 720 Yds with anti-backlash gears, or 0.5 degree or 1800 Yds without anti-backlash gears (see 6.2.1).

FIGURE 4. Single input-triple output, nonisolated channel, 36 to 1/18, 1 and 36 gear ratio type.

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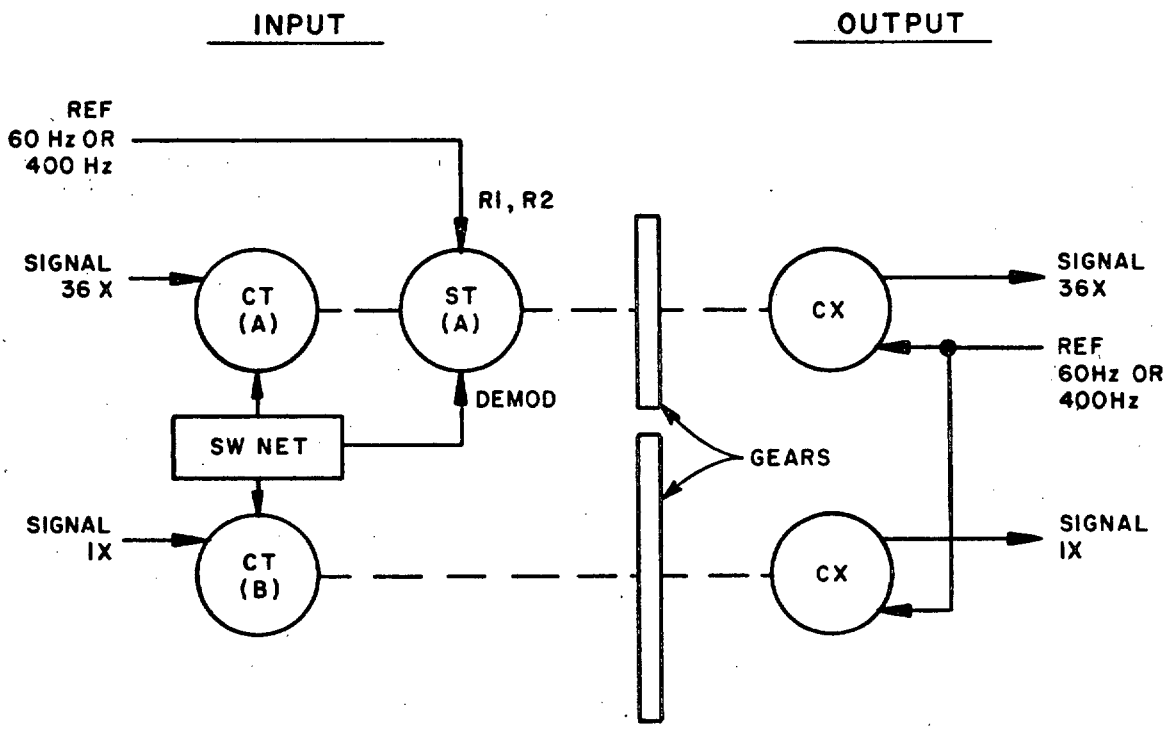
SH 11383

NOTES:

1. ST(A) is the 23ST64a with CT(A). ST rotor leads (R1, R2) and demodulator input leads brought out.
2. CT(B) is a standard CT housed separately.
3. CT(A) and (B) rotor leads connect to input of speed switching network. Output of speed switching network connects to ST demodulator input (DEM).
4. Output synchros may be CX, TX, or TRX, size 23.
5. Converter size is for one panel (see figure 8).
6. 36X equivalent to 648X or 2K Yds/Rev.
   1X equivalent to 18X or 72K Yds/Rev.
   1/18X equivalent to 1X or 1296K Yds/Rev.
7. 36X accuracy = 1 degree or 6 Yds.
   1X accuracy = 0.5 degree or 100 Yds.
   1/18X accuracy = 0.2 degree or 720 Yds with anti-backlash gears, or 0.5 degree or 1800 Yds without anti-backlash gears (see 6.2.1).

FIGURE 5. Double input-triple output, nonisolated channel, 1 and 36 to 1/18, 1 and 36 gear ratio type.

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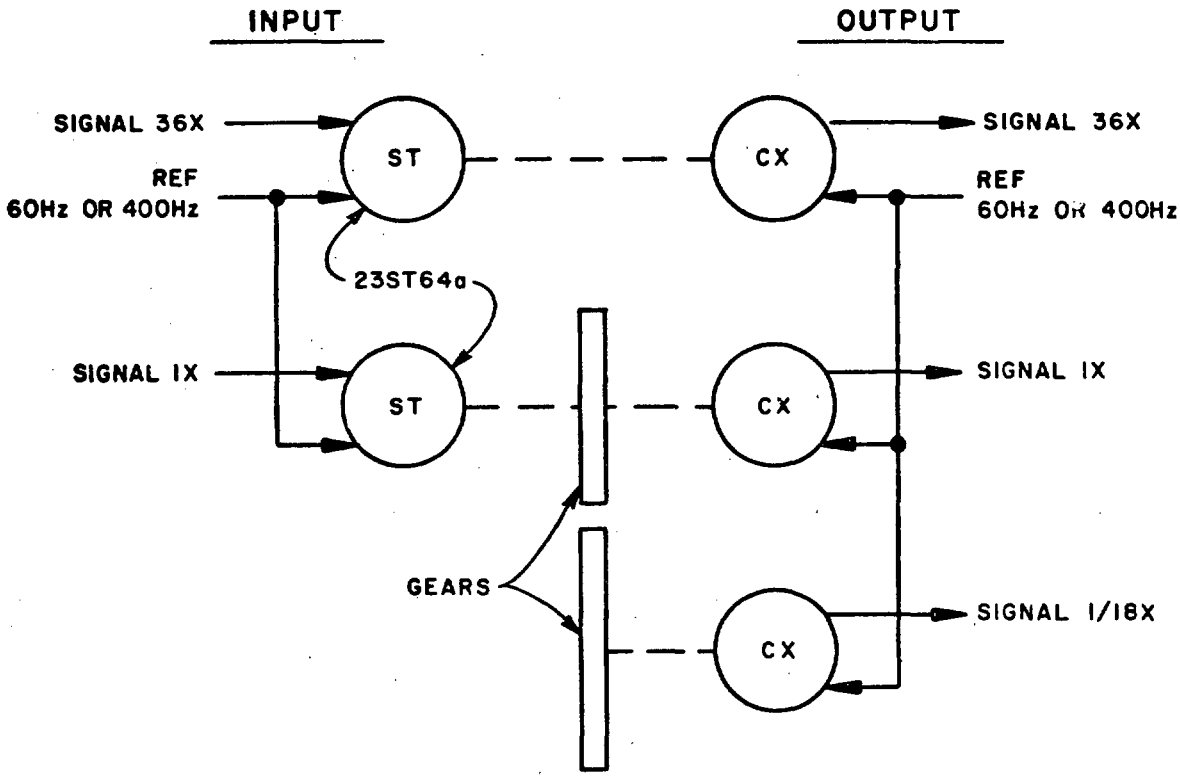
SH 11384

NOTES:

1. ST(A) is the 23ST64a with CT(A). ST rotor leads (R1, R2) and demodulator input leads brought out.
2. CT(B) is a standard CT housed separately.
3. CT(A) and (B) rotor leads connect to input of speed switching network. Output of speed switching network connects to ST demodulator input (DEM).
4. Output synchros may be CX, TX, or TRX, size 23.
5. Converter size is for one panel (see figure 8).
6. 36X accuracy = 1 degree or 6 Yds = 648X or 2K Yds/Rev.
   1X accuracy = 0.5 degree or 100 Yds = 18X or 72K Yds/Rev.

FIGURE 6. Double input-double output, nonisolated channels, 1 and 36 to 1 and 36 gear ratio type.

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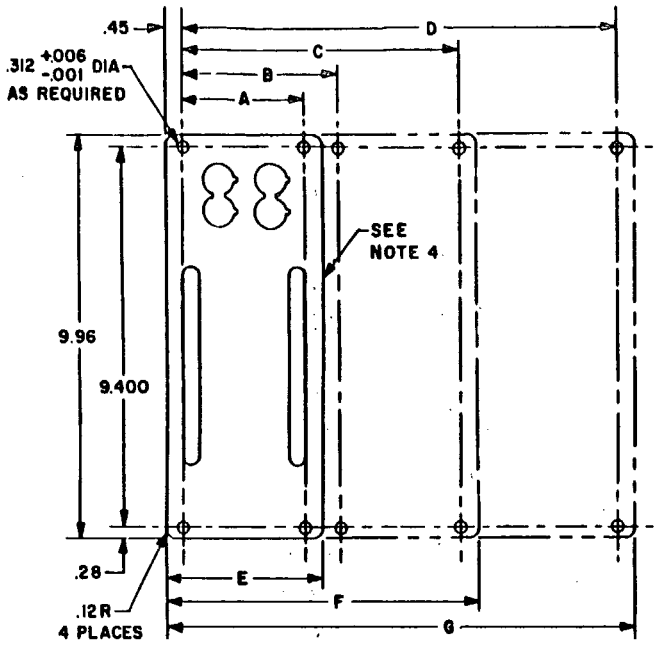
SH 11385

NOTES:

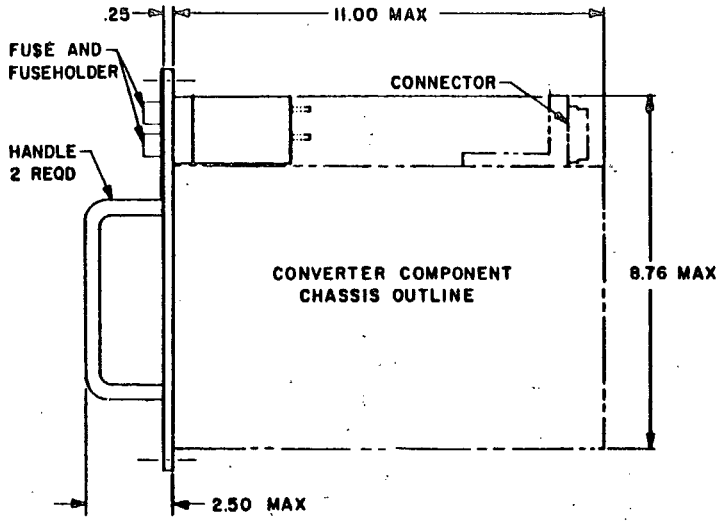
1. Output synchros may be CX, TX, or TRX, size 23.
2. No-speed switching network is provided.
3. Converter size is for one panel (see figure 8).
4. 36X equivalent to 648X or 2K Yds/Rev.
   1X equivalent to 18X or 72K Yds/Rev.
   1/18X equivalent to 1X or 1296K Yds/Rev.
5. 36X accuracy = 0.5 degree or 3 Yds.
   1X accuracy = 1 degree or 200 Yds.
   1/18X accuracy = 0.2 degree or 720 Yds with anti-backlash gears, or 0.5 degree or 1800 Yds without anti-backlash gears (see 6.2.1).

FIGURE 7. Double input-triple output, independent or isolated channels, 1 and 36 to 1/18, 1 and 36 gear ratio type.

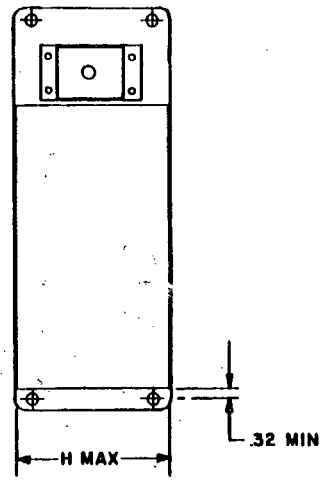
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FRONT VIEW



SIDE VIEW



REAR VIEW

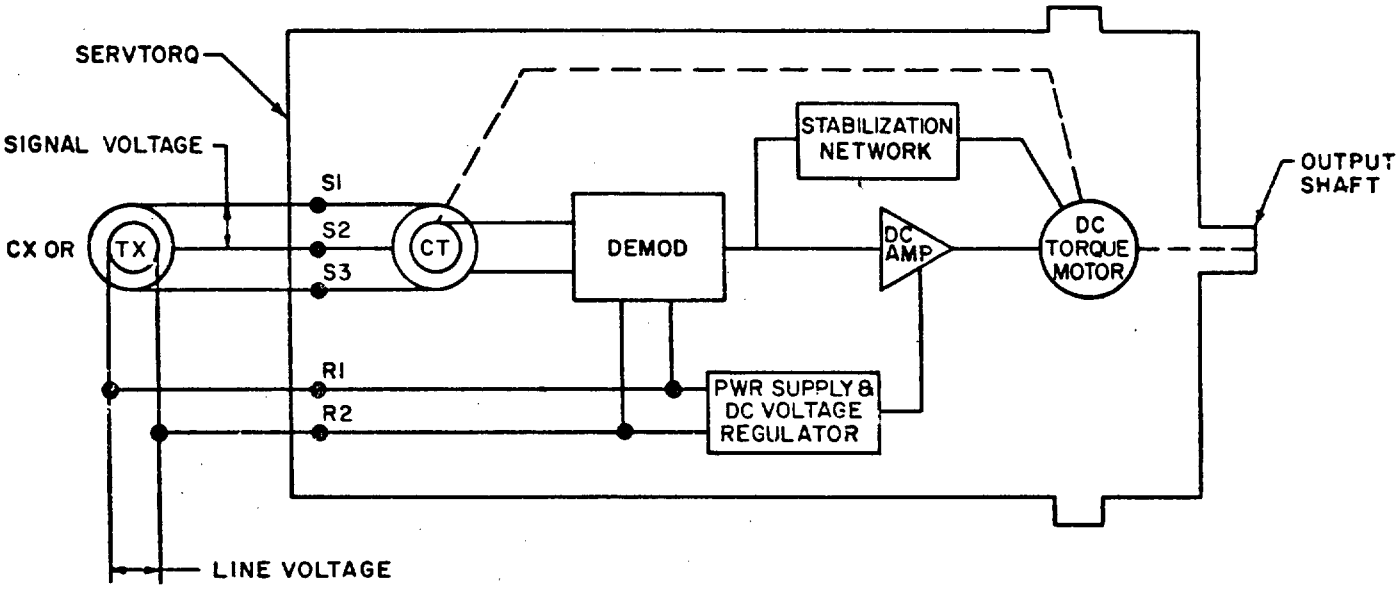
Converter front panel width	Dimensions Mounting holes				Panel width			
	A	B	C	D	E	F	G	H
1 panel	3.060	----	----	----	3.96	----	----	3.88
2 panel	3.060	4.000	7.060	----	----	7.96	----	7.88
3 panel	3.060	4.000	7.060	11.060	----	----	11.96	11.88

NOTES:
1. All dimensions are in inches.
2. Tolerances are 0.XX ± 0.02, 0.XXX ± 0.010.
3. Front panel to be painted in accordance with system 37 of MIL-STD-1303.

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FIGURE 8. Converter envelope dimensioning requirements.

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v_p = Primary of reference voltage: 115 V, 50 to 440 Hz.
v_p = Signal voltage: 90 V, 50 to 440 Hz.

FIGURE 9. Functional block diagram of the SERVTORQ.

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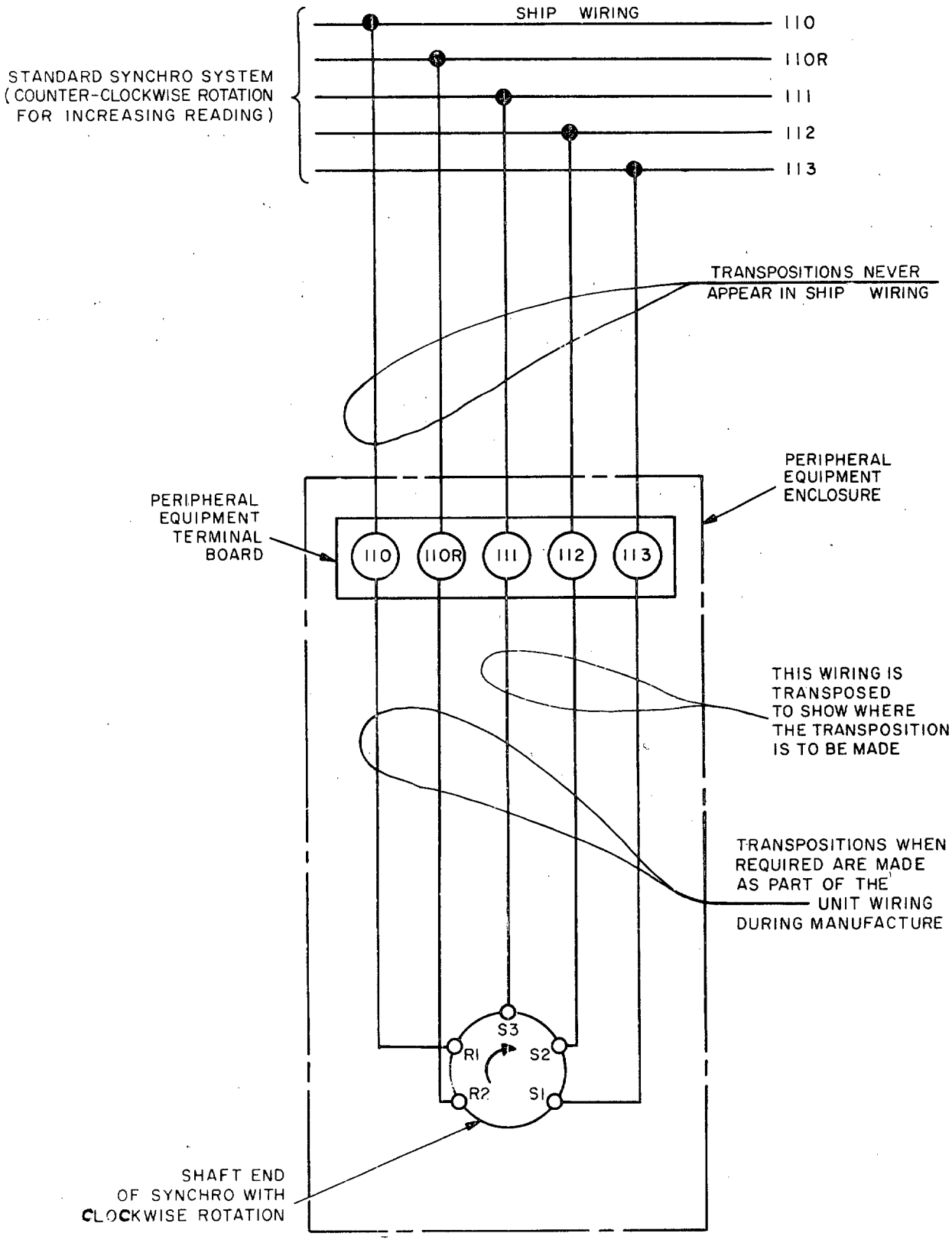
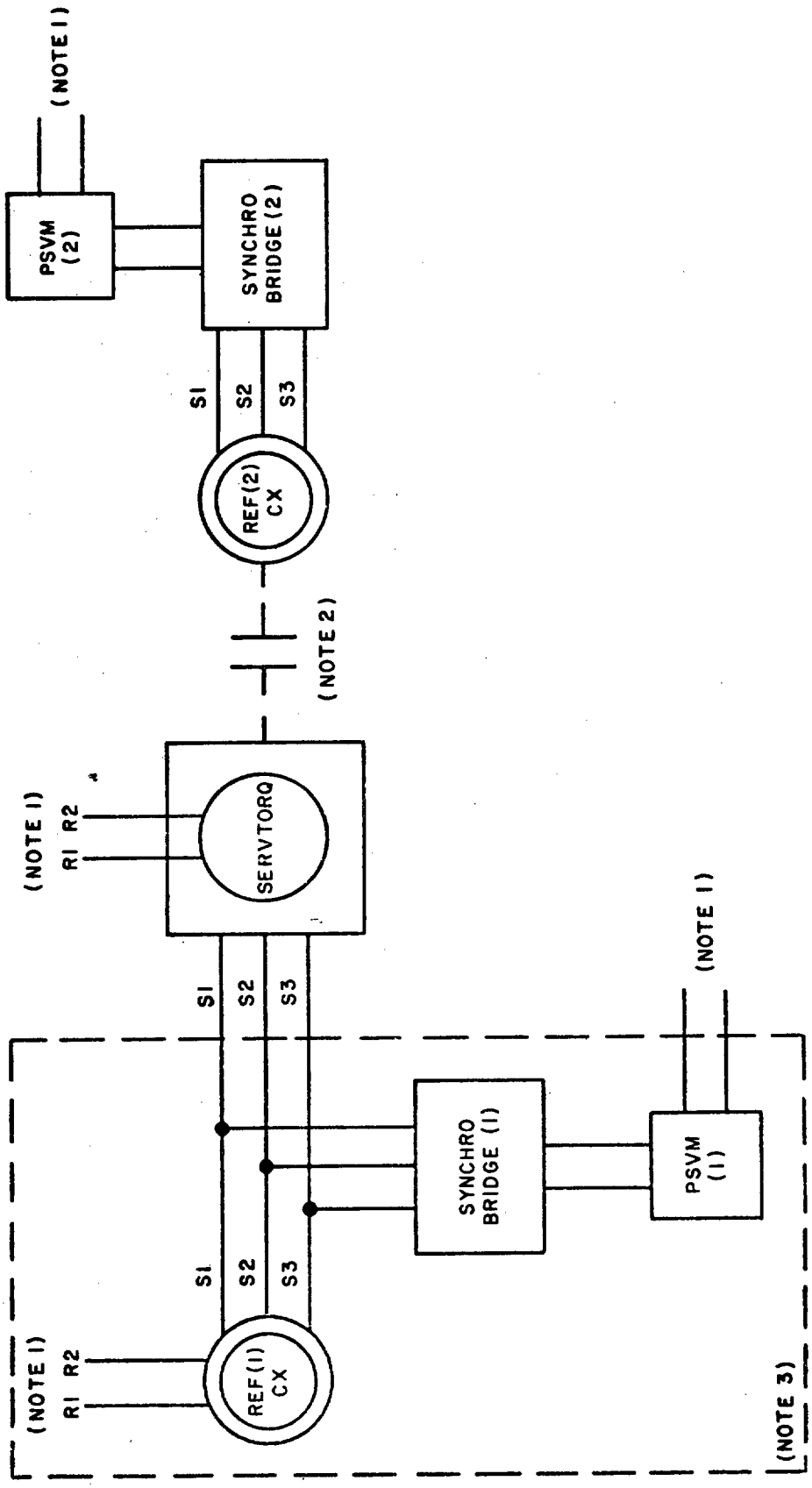


FIGURE 10. Connections for clockwise rotating synchro.

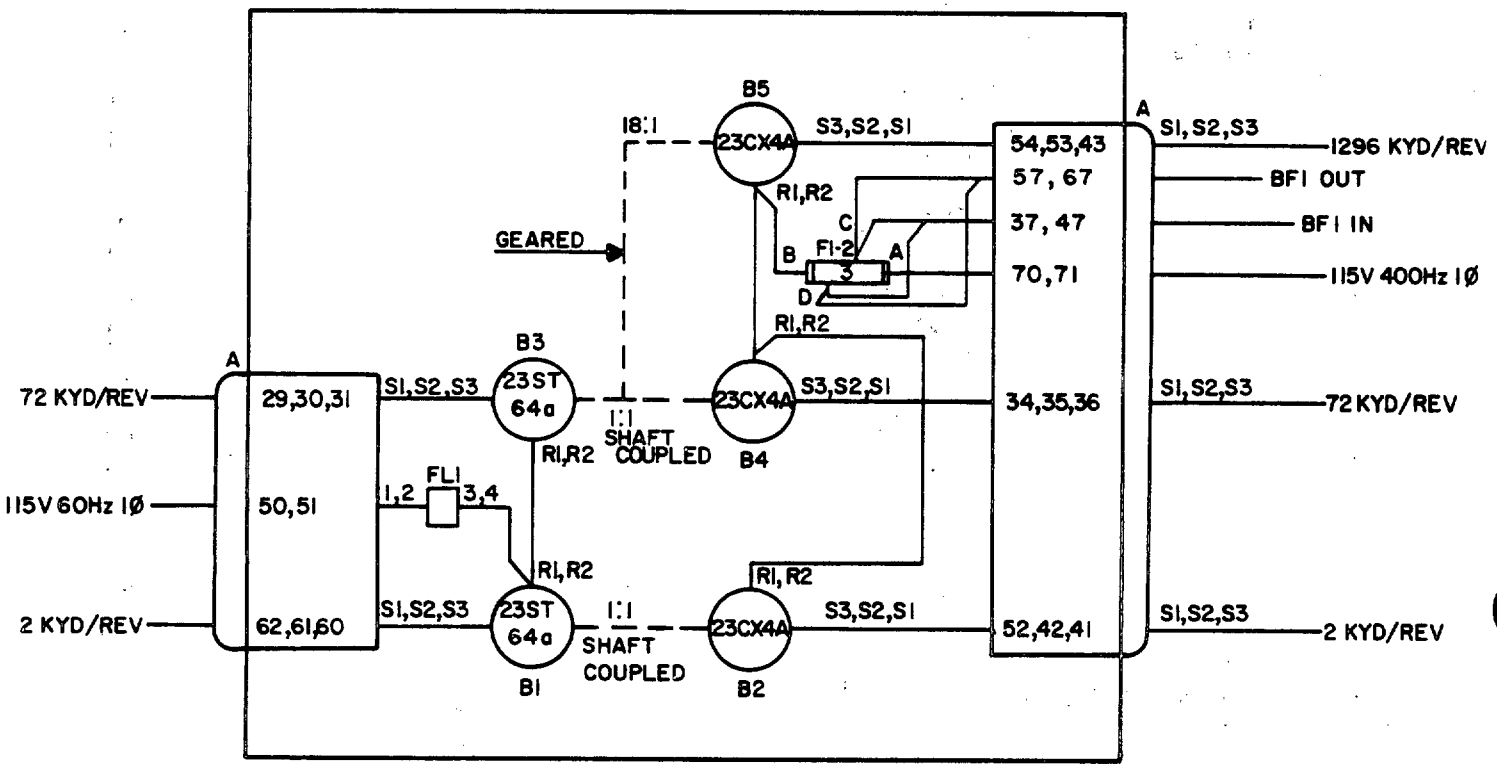
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FIGURE 11. Accuracy test circuit for SERVTORQ or other converters.

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SH 13446

FIGURE 12. Example synchro signal converter wiring diagram, using MIL-C-28748 connector.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-24105C(SH)	CONVERTERS, SYNCHRO SIGNAL (PANEL MOUNTED)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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